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MILITARY SPECIFICATION
TECHNICAL MANUALS: GENERAL PREPARATION OF

This limited coordination military specification has been prepared by the Bureau of Naval Weapons based upon currently available technical information, but it has not been approved for promulgation as a revision of Military Specification MIL-H-5474A. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in procurement.

1. SCOPE

1.1 Scope. This specification covers the general requirements for preparation of technical manuals. Where the requirements of technical content specifications governing these manuals conflict with requirements specified herein, the requirements of the technical content specification shall apply.

1.2 This specification contains the following sections:

1. SCOPE
2. APPLICABLE DOCUMENTS
3. REQUIREMENTS
   3.1 General Requirements
   3.2 Format Requirements
   3.3 Illustration Requirements
   3.4 Reproducible Copy Requirements
4. QUALITY ASSURANCE PROVISIONS
5. PREPARATION FOR DELIVERY
6. NOTES

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-H-7863(Aer)	Handbooks and Breakdowns: Title Pages; "A" Pages and Revised Text Pages; General Preparation of
MIL-N-17792(Aer)	Negatives for Reproduction; General Preparation of

STANDARDS

Military

MIL-STD-12	Abbreviations for Use on Drawings and in Technical Type Manuals
MIL-STD-15	Electrical and Electronic Symbols

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MIL-STD-16	Electrical and Electronic Designations
MIL-STD-17	Mechanical Symbols
MIL-STD-122	Color Code for Chassis Wiring for Electronic Equipment

PUBLICATIONS

Bureau of Naval Weapons

ANA Bulletin No. 143	Specifications and Standards; Use of
ANA Bulletin No. 261	Abbreviations and Contractions for Use on Airborne Equipment; Approved List of
NAVWEPS OP 1700	Volume 1 - Standard Fire Control Volume 2 - Standard Fire Control Symbols; Underwater Related Quantities Volume 3 - Standard Fire Control Symbols; Missile Related Quantities

Government Printing Office

United States Government Printing Office Style Manual

Department of Defense

DD-441 (Attachment)	Industrial Security Manual for Safeguarding Classified Information

(When requesting specifications, standards, NAVWEPS OP's, drawings, and publications refer to both title and number. Copies of this specifications, applicable specifications, and other documents may be obtained upon application to the Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania. Attn: DLI. Copies of DD-441 (Attachment) may be obtained upon application to the nearest Office of Inspector of Naval Material. The U.S. Government Printing Office Style Manual may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 General Requirements

3.1.1 Application. Material for the technical manual shall be furnished in the form (manuscript, reproducible copy, negatives, etc.) as specified in the contract. Unless otherwise stated, reproducible copy shall be furnished. Reproducible copy shall cover only one side of a sheet. Material shall be delivered in final form with all phases prepared for offset printing.

3.1.1.1 Copy shall be typeset or typewritten (produced on office composing machines) whichever is more economical or practicable for the particular piece of work. When applicable and approved by the Procuring Activity, copy may be produced as the result of an Electrical Accounting Machine (EAM) run. Text, other than tabular matter, shall be divided into two vertical columns. Right-hand margins shall be justified when justification is an integral part of the composition process, such as when typecasting, phototypesetting, and typesetting machines or justifying perforated-tape machines are used. Except for headings (marginal copy) or

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special display faces, typeset, typewritten, and EAM copy shall not be combined. (See figures 1 and 2.)

3.1.2 Types of Manuals.

3.1.2.1 Single-Volume. Single-volume manuals are normally used to describe a single component of equipment, a group of related components, or a relatively simple system.

3.1.2.2 Multiple-Volume. Multiple-volume manuals are normally used to describe a weapon system, a highly complicated equipment, or a system (such as an entire aircraft, fire control, missile system or electronic system, etc.). Each volume of a multiple-volume manual issued under a separate binder or cover requires a technical manual identifying number. (See 3.2.2.4.1.)

3.1.2.3 Supplement. A supplement is a subsidiary document which complements information in a manual. Supplementary manuals are normally used for the following:

(a) to add to a commercial manual.

(b) to provide additional data for new models of equipment deemed more expedient or practicable as a supplement than the issuance of a change or revision.

(c) to isolate classified information.

3.1.3 Types of Issues.

3.1.3.1 Basic. The basic issue of a technical manual is the initial edition prepared in full conformance to this specification and in accordance with the requirements specified in the applicable contract.

3.1.3.2 Changes. A change is any alteration in a portion of a manual already in existence, accomplished by replacement, addition, or deletion of pages and/or illustrations, but not large enough to require a complete revision of the manual. Reproducible copy shall be prepared and furnished only for altered or additional pages.

3.1.3.3 Revisions. A revision is a second or subsequent edition of a manual which supersedes the preceding edition.

3.1.3.3.1 When To Prepare a Revision. A revision shall normally be prepared where pages which presently require addition, deletion, or other changes in information plus back-up pages to the current change pages total over 60 percent of the manual. Approval of the Procuring Activity shall be obtained prior to preparing a revision. Manuals consisting of 8 pages or less or which do not include a title page shall be completely revised in each instance instead of changed. (See 3.4.7.)

3.1.4 Coverage of Production Changes. Basic manuals, changes, and revisions shall provide the best possible coverage which can be given at the time of delivery of the manual; delivery shall not be delayed for incorporation of material which can be added by later change or revision. This recognizes that, where equipment is new and undergoing constant change during production, it is not always possible to furnish a basic manual (or initial coverage by change or revision) that will satisfactorily cover all equipment to be delivered throughout the contract period.

3.1.4.1 After delivery of the basic manual (or change or revision) to cover equipment initially, data shall be supplied as required to afford complete coverage of all delivered equipment.

3.1.5 Security Classification. The security classification of the manual shall be furnished by the Procuring Activity as soon as requirements are contractually established. Reproducible copy shall not be prepared prior to the

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receipt of the security classification information. No information of a security classification higher than that established for the manual shall be included. Information of a higher classification considered to be essential shall be brought to the attention of the Procuring Activity. (See 3.4.9.3.3.) In accordance with DD-441 (Attachment) the assigned classification shall be centered at top and bottom of each classified page and its backing page.

3.1.5.1 Security Classification Markings in Text and Figure Captions.

3.1.5.1.1 Markings in Text. Classified paragraphs and subparagraphs shall be marked (C) for Confidential and (S) for Secret, between the paragraph number and heading or start of text if there is no heading. (For example: 2-45. (C) CHECKOUT - or 2-46. (S) The test equipment.....) (See figure 3, sheets 1 and 2.)

3.1.5.1.2 Marking in Figure Captions. When classified illustrations are to be included in the manual, the classification of the illustration shall be indicated by marking (C) or (S) as applicable between the figure number and the title of the illustration. (For example: Figure 2-7. (C) Missile Guidance System.)

3.1.5.2. Markings on Reproducible Pages.

3.1.5.2.1 Reproducible pages of a classified manual shall be marked in accordance with the security classification assigned to their content and on the basis of how they will be assembled in the printed copy of the manual as follows and as shown in figure --:

a. Pages which contain classified information shall carry the assigned classification markings. (See figure 3, sheets 1 and 2.)

b. Pages which contain no classified information or are blank but back or are backed by pages which contain classified information shall carry the marking of the classified page followed by (U). (See figure 3, sheets 3 and 4.)

c. Pages which contain no classified information and back or are backed by pages which contain no classified information shall carry the word UNCLASSIFIED in lieu of the classification marking. (See figure 3, sheets 5 and 6.)

3.1.5.3 Restricted Data. When the designation of restricted data is assigned to a manual in addition to the assigned security classification, the restricted data marking as shown in the latest edition of DD-441 (Attachment) shall appear on each page which includes information in the restricted data category.

3.1.5.4 Classified Supplements. Whenever possible, classified information shall be prepared as a classified supplement to the manual and the manual shall be unclassified. The format and content of the information shall be the same as if prepared for the manual proper. The introduction to the supplement shall have a statement saying in effect, that "This is a supplement to the (title) manual, number (identifying number), prepared in order that the basic manual may be unclassified." Where the manual is to be maintained for the Procuring Activity, the supplement shall be changed and revised in the same manner as a basic manual. Both the supplement and the basic manual shall contain a cross reference note on the respective title pages.

3.1.6 Content. The requirements for the technical content of the manual shall be in accordance with the applicable technical content specification (5) cited in the contractual document. Written matter shall be as simple as is consistent with technical accurancy and information shall be presented logically so

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that it is understandable to all technicians. All related data shall be grouped in a logical manner and shall be so arranged in layout that there is no break in text continuity from page to page except as necessary for related illustrations. Use of notations such as "Continued on page _____" shall not be permitted.

3.1.7 Style. The text shall be factual, concise, free from misinterpretation, and adequate for the intended reader level. It should

(a) Omit basic theory, that is, technical manuals for a specific equipment shall not contain basic theory on types of equipment, but shall include functional description as necessary for understanding and application.

(b) Where possible, avoid phraseology requiring specialized knowledge.

(c) Incorporate needed information from other publications when reference to them is impractical. (See 3.1.13.5.)

(d) Provide publication arrangement information only in front matter, introduction, and index; not throughout the text.

(e) Avoid superfluous words and phrases, such as "the" preceding nomenclature and in procedural steps. Use "because" instead of "for the purpose of," "to" instead of "in order that," and similar phraseology.

(f) Use "preceding" instead of "above" and "following" instead of "below" when referring to text position.

(g) Provide descriptive headings for sections, paragraphs, and other subdivisions, as appropriate to pinpoint location of specific information. Avoid identical and "general" headings.

(h) Emphasize meaning more closely by using "when" or "where" instead of "if" provided meaning is not impaired.

3.1.7.1 Capitalization and Punctuation. The United States Government Printing Office Style Manual shall be used generally as a guide for capitalization, punctuation, and compound word forms. In addition, the following shall apply:

(a) The word "drawing" preceding a drawing number shall be abbreviated (see 3.1.10) and shall have an initial capital letter only in headings, captions, and the first word of a sentence. (For example: dwg ---.)

(b) The first letter of each word of official (government) nomenclature shall be capitalized including Mark and Mod. (For abbreviations of Mark and Mod, see 3.1.10.)

(c) When mark and mod are common nouns, they are not capitalized.

(d) Reference designations are not punctuated. (For example: Remove tube V101 and capacitor C19.)

(e) Quotation marks shall be avoided, especially for capital letters, designations, and emphasized words and phrases. (For example: use O-ring, not "O" ring.)

(f) Omit commas between hyphenated phrases. (For example: 115-volt 400-cycle 3-phase ship power.)

3.1.7.2 Spelling. The United States Government Printing Office Style Manual shall be the guide for spelling. Words not in the style manual shall be spelled in accordance with Webster's New International Dictionary.

3.1.7.2.1 The following exceptions shall prevail:

(a) fuse: for electrical circuits
(b) fuze: for explosive trains

3.1.7.2.2 For inanimate objects the possessive case shall not be used in its apostrophized form. (For example, use "ship course" not "ship's course".) This complies with definitions for fire control symbols as described in NAVWEPS OP 1700.

3.1.7.2.3 The following word combinations, in all their forms, shall be spelled without hyphens (except when

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reference is to an equipment inscription, in which case the inscribed form shall be used).

(a) high voltage
(b) low voltage
(c) motor generator

3.1.8 Grammatical Person and Mode. The second person imperative shall be used for operational procedures. (For example, "Break casing bead loose from wheel flange".) The third person indicative shall be used for description and discussion. (For example, "The torsion link assembly transmits torsional loads from the axle to the shock strut".)

3.1.9 Nomenclature. Nomenclature consistency shall be maintained. For example, a part identified as a "cover" shall not be identified thereafter as a "plate". Portions of nomenclature used in the text shall agree with nomenclature on related engineering and design drawings.

3.1.9.1 Official nomenclature assigned by the Government shall be used in its normal reading sequence. Transposed nomenclature is to be used only in illustrated parts breakdowns (parts lists) and indexes. For example:

(a) text: Parallax Corrector Mk 4
(b) parts list: Corrector, Parallax, Mk 4

3.1.9.2 When nomenclature is other than official nomenclature, lower-case letters shall be used. For example:

(a) parallax corrector
(b) radar set

3.1.9.3 When two or more items have the same nomenclature, modifiers shall be added to define clearly the function of each item, its distinctive physical properties, or both.

3.1.9.4 The first time an equipment item appears it shall be identified fully. Further references in the same chapter or section may omit official nomenclature modifiers, where there is no possibility of confusion. For example: Missile Launcher Tube Mk 1 Mod 0 introduces the equipment by its official nomenclature. Thereafter the mark and mod designations may be omitted, unless required for differentiation between other marks and mods of the tube. Subsequent references may be to "launcher tube" or to "tube", depending upon need for modifiers as determined by context. If "missile" is used as a modifier, "launcher" must be used also. The item is not a "missile tube", it is a "missile launcher tube".

3.1.9.5 Where more than one mark or mod of equipment is described, the plural shall be indicated as follows:

(a) Mines Mk 6 and Mk 8
(b) Mine Mks 101, 102, 107, and 109
(c) Radar Sets AN/SPG-5 and AN/SPG-6
(d) Mine Mk 6 Mods 0 and 1

3.1.10 Abbreviations. The use of abbreviations shall be held to a minimum. However, abbreviations familiar to the user are permitted. Any abbreviations used shall be in accordance with Standard MIL-STD-12 and ANA Bulletin 261. In case of conflict Standard MIL-STD-12 shall govern. When used as official nomenclature assigned by the Government, Mk and Mod shall be abbreviated with initial capital letter and no period.

3.1.11 Graphic Symbols. Graphic symbols for electric and electronic parts shall be in accordance with Standard MIL-STD-15. Graphic symbols for mechanical parts shall be in accordance with Standard MIL-STD-17.

3.1.12 Reference Designations. Reference designations for electrical and electronic symbols shall be in accordance with Standard MIL-STD-16.

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3.1.13 References.

3.1.13.1 Model or Type References. Text shall refer only to models or types of equipment covered by the technical manual. To facilitate later incorporation of additional models or types, reference to model or type designation shall be held to a minimum consistent with clarity. Any such references made shall be expressed in definite terms such as model or type designation, part number, serial number range, or similar means. Such terminology as "on early serial numbers", "on some late models or types", etc., is not acceptable.

3.1.13.2 Figure References. Where reference is made to figures, the reference shall be to the figure number normally set within parentheses, for example, "(See figure 3-6.)." The page number shall not be used. (See figure 3, sheets 7 and 8.) When reference is made to items shown on figures by index numbers (or reference symbols), the numbers shall be indicated in the following manner: "The ON-OFF switch (34, figure 3-6) on the co-pilot's instrument panel controls the IFF radio."

3.1.13.3 Multiple References. Where references are made in a paragraph to several items in the same figure, as in instructions detailing work procedures, the figure number need be given only once at the beginning of the reference (usually the beginning of the paragraph) with index numbers (or reference designators) of the items placed in parentheses in the body of the text where pertinent. Care shall be taken that such references are entirely clear. An example is the following portion of a paragraph:

"3-10. DISASSEMBLY OF AIR VALVE (See Figure 3-4)

1. Unscrew safety ring retainer (2) from valve body (1) and remove safety ring washer (4)."

3.1.13.4 Paragraph References. In reference to text matter placed elsewhere in the manual, the reference shall be to paragraph numbers only as "paragraphs _____ through _____." Page numbers shall not be included.

3.1.13.5 References to Other Publications. Reference shall be made to other publications such as to another manual for operation, maintenance, or overhaul procedures when specifically permitted by the applicable technical content specification. Reference to specific paragraphs or pages in other publications shall not be made. If the reference is to the entire contents or a major portion of another publication, the cross reference shall be only to the publication identifying number. When a small amount of information is needed, the applicable portion shall be extracted, condensed, modified as necessary, and included in the technical manual. Essential references to other than Bureau of Naval Weapons publications, drawings, etc. must include the identity of the issuing activity.

3.1.13.6 Number References. The United States Government Printing Office Style Manual shall be used as a guide for numbers. Double statement of numbers (a number spelled out followed by a numeral in parentheses) shall not be used. Numbers less than unity involving decimals shall begin with a zero before the decimal point.

3.1.13.7 Specifications and Standards References. When Government specification or standard numbers are referred to, only the basic number shall be used, omitting the revision letter-suffix unless it is essential to mention a particular issue. Reference shall be as follows: Federal Specification P-S-661, Specification MIL-C-5020, Standard MIL-STD-105, etc.

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3.1.13.8 Material References. All material referred to, such as lubricants, sealing materials, abrasives, etc, shall be identified by Government specification numbers, wherever applicable. The class of specification called for (Federal, Military, Air Force-Navy, Army, Navy, Air Force, non-Governmental, etc.) shall be in accordance with the order of precedence given in ANA Bulletin No. 143.

3.1.13.8.1 When the contractor does not know equivalent specifications for one of the Services, he shall request this information from the Procuring Activity, listing complete information concerning its composition, properties, characteristics, application, etc.

3.1.13.8.2 Where design considerations require the use of a proprietary (non-Service) specification, a standard (Service) specification shall also be listed as an emergency alternate, if such an alternate specification exists.

3.1.13.8.3 Identifications of materials should, wherever practicable, be in the form of a table with additional information as to points of application as required.

3.1.13.9 Temperature References. Temperature readings shall be given in degrees as indicated in the applicable technical content specification. Where the technical content specification does not give these instructions, the temperature references shall be in the unit of measurement in which the equipment is calibrated.

3.1.13.10 Liquid Measure References. All references to liquid capacities shall be given in U. S. Standard units of liquid measure. When scientific or medical equipment is of a type which makes references to metric weights and measures customary, such references may be used.

3.1.13.11 Reference to Health Hazards Precaution Data. When hazardous chemicals or adverse health factors in the environment or use of the equipment cannot be eliminated, appropriate precautionary requirements shall be included. Necessary protective devices for personnel shall be indicated.

3.1.13.12 Speed and Distance References. Refer to speed and distance as calibrated by the equipment concerned.

3.1.13.13 Table References. When text matter is presented in tabular form for reference purposes, reference is made to the table number. The page number shall not be used.

3.1.13.14 References to Equipment Markings. References to switch positions and panel markings shall be referred to exactly as marked on the equipment (spelling, abbreviations, capitalization, etc.). Where necessary, explanations of these markings may be included.

3.1.13.15 Signs and Symbols. Word symbols such as /, -, ±, o may be used when preceding or followed by values. In tabular matter, the symbol for percent may be used when preceded by a numeral. In text, percent shall be spelled out. When used, signs and symbols shall be as listed in the United States Government Printing Office Style Manual. Fire control symbols shall be in accordance with NAVWEPS OP 1700. Chemical symbols should not be used. In lieu of symbols, use words such as carbon dioxide, hydrogen peroxide, and so forth.

3.1.14 Notes, Cautions, and Warnings. These adjuncts to the text shall be used as defined, and in conformance with the requirements of figure 1. These requirements are illustrated in figure 3, sheets 5 through 8. An alternate method

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for display of cautions and warnings is shown in figure 2. The method selected shall be used consistently throughout a manual. They shall be used sparingly consistent with real need. Notes follow the applicable text. Cautions and warnings precede the instructions for the procedure or practice involved.

NOTE: An essential highlight for an operating procedure, condition, or arrangement required for clarity and emphasis.

CAUTION

An emphatic notice requiring correct operating or maintenance procedures or practices to prevent damage to or destruction of equipment, and warning against improper procedures and practices. A major caution warns against a procedure, condition, or practice which may destroy the effectiveness of the equipment of ship (mission).

WARNING

A more emphatic notice requiring correct operating or maintenance procedures and practices and ensuring safe conditions, to prevent personal injury or loss of life.

3.1.15 Footnotes. Footnote references in the text shall be used sparingly. If used, consecutive numerals beginning with "1" shall be used for footnote references in the text. In tables, footnote references shall be numbered separately for each table, in accordance with the United States Government Printing Office Style Manual. Footnotes to the text shall be placed at the bottom of the page; footnotes to tables below each table. (See figure 3, sheets 1 and 2.) Footnotes in the text shall not be used for mandatory requirements. Within a multivolume series, each volume shall have its own set of footnotes.

3.2 Format Requirements.

3.2.1 Arrangement. The arrangement of the manual into chapters and sections shall be as required by the applicable technical content specification. The word "section" as used throughout the remainder of this specification shall be construed to mean either section or chapter (whichever is required by the technical content specification). A section shall consist of one or more main paragraphs, with each divided into subparagraphs as necessary.

3.2.1.1 Space Conservation. Layout shall conserve space without lessening usability or clarity of material. Blank pages and spaces shall be avoided whereever possible.

3.2.1.2 Component Parts. The technical manual shall contain the following component parts as applicable:

- Front Matter
  - Title Page or Title Panel
  - List of Effective Pages ("A" Page)
  - Flyleaf
  - Foreword
  - Table of Contents
  - List of Illustrations
  - List of Tables
  - Safety Summary
- Sections
- Appendix
- Index

3.2.1.3 Title Page. A title page (except as indicated in 3.2.1.4) shall be furnished as part of the reproducible copy for each basic manual, change, revision, and supplement, and shall be prepared in accordance with the requirements of Specification MIL-H-7863(Aer).

3.2.1.4 Title Panel. A title panel in lieu of a title page shall be furnished when required by the technical content specification.

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3.2.1.5 List of Effective Pages ("A" Page). The reverse side of the title page shall contain the list of effective pages ("A" page) and shall be prepared in accordance with the requirements of Specification MIL-H-7863(Aer).

3.2.1.6 Flyleaf. If the requirement for a flyleaf is specified in technical content specifications, it shall be a page inserted immediately following the "A" ("B", "C", "D", etc.) page. The purpose of the flyleaf shall be to contain a listing of interim changes issued for a manual with indication of their incorporation status. The word "Flyleaf" shall appear in the lower outside corner of the page. The basic flyleaf page shall not be numbered, but ensuing pages shall be numbered "Flyleaf -2", "Flyleaf-3", "Flyleaf-4", etc. The flyleaf shall be permanent and remain as part of the technical manual history.

3.2.1.7 Foreword. If the requirement for a foreword is specified in technical content specifications, it shall be a page inserted immediately preceding the table of contents. The word "Foreword" shall appear in the lower outside corner of the page. (See figure 4.) The basic foreword page shall not be numbered, but ensuing pages shall be numbered "Foreword-2", "Foreword-3".

3.2.1.8 Table of Contents. The requirements for preparation of the table of contents shall be in accordance with the applicable technical content specification. The table of contents shall list numbers and titles of sections and first, second, and third order paragraph heads with their initial page numbers. Single or double column format may be used; double is preferred. Layout shall conform to the following example:

TABLE OF CONTENTS

Section		Page
I.	INTRODUCTION	1-1
1-1	Description	1-1
1-2	Body	1-1
1-3	Wing	1-2
1-4	Armament	1-3
II.	ELECTRICAL POWER SUPPLY AND DISTRIBUTION SYSTEM	2-1
2-1	Special Tools	2-1
2-2	Electrical Power Loading Data	2-1

3.2.1.9 List of Illustrations. When more than five illustrations are used in any technical manual or any one volume of a multiple volume manual, a list of illustrations shall be shown. The list of illustrations shall show the number, title, and page number of each illustration. Listings shall be in consecutive order of the illustrations (figures). Single or double column format may be used; double is preferred. Layout shall conform to the following example:

MANUAL NO. List of Illustrations

LIST OF ILLUSTRATIONS

Figure		Page
1-1.	Electronic Components of Intercommunication Set AN/AIC-10	1-1
1-2.	Intercommunication Set AN/AIC-10, Electronic Components	1-2
2-16.	Audio Frequency Amplifier AM-843/AIC-10, Connection Diagram	2-14
2-17.	Dynamotor DY-76/AIC-10 and DY-77/AIC-10, Connection Diagram	2-14

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3.2.1.10 List of Tables. When there are more than five tables in the manual, the requirements for the list of tables are the same as for the list of illustrations. The list of tables shall follow directly after the list of illustrations in the following format:

LIST OF TABLES

Table	Page

Table	Page

3.2.1.11 List of Charts and Graphs. When charts and graphs are used in the manual, they shall be listed by the assigned numbers and shall be shown in the list of illustrations.

3.2.1.12 Safety Summary. When applicable, the safety summary shall list all general type precautions that are not pinpointed to specific text paragraphs, such as high voltage warnings, warnings against servicing alone, and a resuscitation statement. In addition, the WARNINGS and the major CAUTIONS that appear throughout the text should be listed in their text sequence. These WARNINGS and major CAUTIONS should be rephrased if necessary so that they are completely meaningful when presented out of context. At the end of each WARNING and major CAUTION the number of the text page on which each appears should be indicated. If a WARNING or major CAUTION is repeated in the text, it should be listed in the safety summary only once with multiple page references. A major CAUTION is defined as an operating procedure or practice which, if not strictly observed, may destroy the effectiveness of the equipment or ship. (See figure 5, sheets 1 and 2.)

3.2.1.13 Sections or Chapters. The arrangement of the manual into sections or chapters shall be as required by the applicable technical content specification of the Procuring Activity. (See figure 3, sheets 1, 2, and 5 through 8.)

3.2.1.14 Appendix. When required by the applicable technical content specification, or when approved by the Procuring Activity, an appendix may be used to include material not properly falling within any section of the text. The appendix shall not be used in lieu of a change. Pages, paragraphs, and illustrations within an appendix shall be identified by the letter identifying the appendix, followed by an Arabic numeral indicating the page, paragraph, or illustration sequence within the appendix.

3.2.1.15 Glossary. Technical terms and/or abbreviations used throughout a technical manual may be shown in a glossary immediately following the last section and when used shall be the first appendix of a manual.

3.2.1.16 Index. An alphabetic index shall be included when required by the technical content specification and shall be the last portion of the manual. As a minimum, the index shall list all chapter, section, and table titles and all paragraph headings, rearranged as necessary so that the equipment or component is the main indexed entry. For example: a paragraph entitled "Removal of Cover" shall be indexed as "Cover, Removal". Paragraph, figure, and table numbers shall be indicated. One subtopic may be indented to avoid repetition. Subsequent indications shall be avoided. It shall be entitled "Index" and each page shall show the word "Index" in the upper outside corner, plus the first and last items appearing on the page. It shall not be assigned an appendix or

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section number. Pages shall be numbered Index-1, Index-2, etc.

3.2.2 Format/Make-Up.

3.2.2.1 Composition Methods.

3.2.2.1.1 Typeset Copy Preparation. When the contract requires one reproducible copy suitable for photographic reproduction, each page of the manual shall be furnished in reproduction size on sheets of white enamel coated paper or a suitable grade of proof paper. (See figure 6.)

3.2.2.1.1.1 Type faces and details of preparation shall be in accordance with figure 1, sheets 1 through 6. Other type faces may be used if they correspond to faces approved for typeset copy listed in figure 1, sheets 1 through 5. A sample sheet identifying and showing any other type faces proposed for use shall be furnished the Procuring Activity for approval.

3.2.2.1.2 Typewritten Copy Preparation. One reproducible copy suitable for photographic reproduction for each page shall be furnished, typed in single-spaced lines on good white bond paper, using a carbon paper ribbon. The following typefaces, type sizes, and sheet sizes are acceptable (any other mechanical means of producing comparable type faces may be used):

Type Faces	Type Size	Sheet Size
IBM Electromatic (or equivalent)		
Bold Face	12 point	25 percent oversize (figure 9)
Bold Face No. 2	10 point	Reproduction (figure 6)
Modern	10 point	10 percent oversize (figure 7)
Varityper		
250-10-12	12 point	25 percent oversize (figure 9)
D.S.J. (Differential Spacing Justifier)	10 point	Reproduction (figure 6)
Typewriter Face		
10 pitch	Pica	25 percent oversize (figure 9)
12 pitch	Elite	20 percent oversize (figure 8)

3.2.2.1.2.1 Details of preparation shall be in accordance with figure 1, sheets 1 through 6; figure 10, sheets 1 through 4; and figure 11, sheets 1 through 4. Text shall be clean and black without strike-overs or erasures.

3.2.2.2 Reproducible Text Pages. Reproducible text pages (typeset or typewritten) are not acceptable if headings, changes, or corrections are pasted on. If such are necessary, they shall be stripped in by the mortising method. Where full page reproducible copy is prepared in galley form by cold type composition, such copy shall be placed in oversized sheets or properly sized sheets by means of the hot mounting process or stripped in by the mortising process. If the mortising process is used, white opaque tape shall be applied to the backs of the sheets.

3.2.2.3 Negatives. When specified in the contract, first class photo-lithographic negatives, completely opaqued

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and in such condition that no additional work shall be required prior to printing, shall be furnished. *Requirements for* negatives shall be in accordance with Specification MIL-N-17792(Aer).

3.2.2.4 Numbering.

3.2.2.4.1 Technical Manual Number. The contractor shall request the technical manual identifying number from the Procuring Activity as soon as the requirements are contractually established. The assigned number shall be shown on each page. (See figure 11.)

3.2.2.4.1.1 Technical Manual Number *for Supplement.* The identifying number for the supplement shall be obtained from the Procuring Activity. The assigned number shall be shown on each page. (See figure 11.)

3.2.2.4.2 Chapter Numbers. When required by the technical content specification, chapters throughout the manual shall be numbered consecutively in Arabic numerals. Chapter numbers in a supplement shall be the same as those in the manual being supplemented.

3.2.2.4.3 Section Numbers.

3.2.2.4.3.1 When the technical content specification requires chapters and sections, section numbers shall consist of two Arabic numerals separated by a dash. The number preceding the dash will designate the chapter number and the number following the dash will indicate the number of the section within the chapter.

3.2.2.4.3.2 When the technical content specification requires only sections, the sections shall be numbered consecutively in capital Roman numerals.

3.2.2.4.3.3 Section numbers in a supplement shall be the same as those in the manual being supplemented.

3.2.2.4.4 Page Numbers. Except as otherwise specified in 3.2.2.4.4.1, pages shall be numbered on the lower outside corner. Commencing with the first right-hand page of the first section, pages shall be numbered in Arabic numerals separately and consecutively within each section.

3.2.2.4.4.1 Blank Page Numbers. A page number is not required on blank pages since reproducible copy is not required; however a blank page shall be a part of the sequence of page numbers; for example, if page 2-11 is a text page and 2-12 is blank, the following right-hand text page would be page 2-13. For blank pages in classified manuals, see 3.1.5.2.1.

3.2.2.4.4.2 Page Numbers Preceding the First Section. Pages preceding the first section shall be numbered consecutively with lower case Roman numerals (i, ii, iii, iv, etc.). The table of contents shall begin on page i (lower case Roman). If the table of contents is one third of a page or less in depth, it may be placed on page 1-1 (Arabic) above the section title. When required by the technical content specification, the frontispiece shall be the last even-numbered lower case Roman numeral page.

3.2.2.4.4.3 Text Page Numbers. Commencing with the first right-hand page of the first section, pages shall be numbered separately and consecutively within each section in Arabic numerals. (For example: 1-1, 1-2, 2-1, 2-2, etc.) Where it is mechanically more feasible and in manuals of 8 pages or less, the text pages shall be numbered consecutively in Arabic numerals.

3.2.2.4.4.4 Fold-Over Page Numbers. Fold-over pages shall be assigned one

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page number.(See figure 12.) If fold-over pages are backed up, one page number shall be shown on each side of the page. If the reverse side of the fold-over page is blank, both page numbers shall be shown on the face in the lower outside corner.

3.2.2.4.4.5 Appendix Page Numbers. Pages within an appendix shall be numbered by the letter identifying the appendix, followed by an Arabic numeral. (For example, A-1, A-2, etc.)

3.2.2.4.4.6 Index Page Numbers. Pages in the index shall be numbered "Index-1", "Index-2", etc.

3.2.2.4.4.7 Page Numbers in Changes. Change pages shall replace pages of the same numbers, if possible. Added pages which are to be inserted between previously printed pages shall be numbered with the number of the left-hand page (even numbered) only, and suffixed with a capital letter for each added page, starting with "A". (For example, 2-22A, 2-34A, 2-34B, etc.) If it is necessary to insert new pages between 2-34A and 2-34B, such pages shall be numbered beginning with 2-34B, 2-34B-1, etc., and the original page 2-34B shall be renumbered in sequence. The letters "I" and "O" shall not be used. If necessary to go beyond the letter "Z", the contractor should consider and recommend to the Procuring Activity the feasibility of preparing a revision.

3.2.2.4.4.8 Deleted Page Numbers in Changes. When page numbering continuity is broken by a deletion of a page by a change, such page shall be furnished blank except for a centrally located notation, with proper page number indicated as follows: "Page_____, Figure(s)_____, deleted". Deleted pages shall also show the manual identifying number and classification (if any). If however, two deleted pages back each other, or fall in sequence of more than two pages, neither shall be furnished, but a statement such as "Pages_____, Figures_____, deleted", indicating their deletion, shall be placed at the bottom of the preceding page or the top of the following page, to show reason for the break in continuity. Change symbols shall be used in the margin adjacent to such statements.

3.2.2.4.5 Paragraphs.

3.2.2.4.5.1 Basic System. Paragraphs shall be numbered consecutively within each section, using Arabic numerals separated by a dash. The number preceding the dash shall indicate the section and the number following the dash shall indicate the number of the paragraph within the section. (For example, "2-17" would be the 17th paragraph of the second section.)

3.2.2.4.5.2 Supplement Paragraph Numbers. Paragraphs in the supplement shall be numbered consecutively in the same manner as the basic manual.

3.2.2.4.5.3 Appendix Paragraph Numbers. Paragraph numbering within the appendix shall be in the same manner as the basic manual with a capital letter corresponding in alphabetical position to the number of the appendix and shall be used preceding the dash. (For example, "B-14" would be the 14th paragraph of Appendix B.)

3.2.2.4.5.4 Paragraph Numbers in Changes. Paragraphs inserted by a change shall be given the number of the preceding paragraph distinguished by the addition of capital letters starting with "A". (For example, two paragraphs inserted between existing paragraphs 4-29 and 4-30 would be numbered 4-29A and 4-29B.) If it is necessary to insert new paragraphs between 4-29A and 4-29B, such paragraphs would be

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numbered 4-29A-1, 4-29A-2, etc. When paragraph numbering continuity is broken by deletion of text, the paragraph number shall be shown, followed by the word "Deleted", shown in parentheses without quotes.

3.2.2.4.5.5 Unnumbered Paragraphs. Unnumbered paragraphs may be used only when specified by a technical content specification.

3.2.2.4.5.6 Numbering of Procedural Steps. Steps of procedure, such as steps of assembly or disassembly, shall be assigned consecutive Arabic numerals. (See figure 11, sheet 3.) Steps inserted by change shall be given the number assigned the preceding step suffixed by a capital letter. For example, a new step inserted between existing steps 1 and 2 would be numbered 1-A. The same procedure shall be followed in lists of equipment or other lists forming part of the text.

3.2.2.4.6 Notes, Cautions, and Warnings. Notes, cautions, and warnings shall not be numbered.

3.2.2.4.7 Figures.

3.2.2.4.7.1 Requirements. Illustrations, graphs, and charts shall be referred to within the technical manual as figures. They shall be identified by a figure number and a title. Changed illustrations shall carry the same figure numbers as the original illustration. Whenever possible, figure numbers and titles shall be positioned immediately beneath the illustration as it is held in reading position. Illustrations (figures) requiring more than one page shall be identified in the following manner:

Figure 1-1 (Sheet 1). Lubrication Chart.
Figure 1-1 (Sheet 2). Lubrication Chart.

3.2.2.4.7.2 Figure Numbers in Technical Manuals and Supplements. Figure numbers shall be preceded by the word "Figure" and shall consist of two Arabic numbers separated by a dash. The number preceding the dash shall designate the section and the number following the dash shall indicate the number of the illustration (figure) within the section. (For example, "Figure 1-3" would designate the third illustration in the first section.)

3.2.2.4.7.3 Figure Numbering. Figures shall be numbered consecutively in the order of their appearance within the section, appendix, or supplement.

3.2.2.4.7.4 Figures Preceding the First Section. Illustrations (figures) preceding the first section and pertaining to the entire technical manual or to more than one section shall be given the first section figure numbers.

3.2.2.4.7.5 Figure Numbers in the Appendix. Figure numbers in the appendix shall be the same as in the manual except that the letters A or B (Appendix A or Appendix B) shall precede the dash. (For example, "Figure A-1" would indicate the first figure in Appendix A.)

3.2.2.4.7.6 Figure Numbers in Illustrated Parts Breakdown. In an illustrated parts breakdown to be published as a separate manual, the use of a single Arabic number shall apply and shall be used consecutively throughout the breakdown in order of appearance. When the illustrated parts breakdown is a section of a manual, the numbering shall follow the same format used for the manual.

3.2.2.4.7.7 Figures Added by Change. Illustrations inserted with a change and located between two existing illustrations shall have the same number as the preceding illustration, but the number shall be followed by a capital "A" for the first figure, "B" for the second, etc. (For example, Figure 2-5A, Figure 2-5B, etc.)

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(See figure 17.) Illustrations inserted with a change after the last existing illustration in any section shall have the next numbers in sequence.

3.2.2.4.7.8 Deleted Figures. When change or revision material requires deletion of an illustration without substitution of another, the following sentence shall be inserted: "Figure _____ deleted." The same requirement applies to tables.

3.2.2.4.7.9 Table Numbering. Tables shall carry the title and table number at the top of the table, and shall follow the same system of numbering as used for figures, except that the numbers shall be preceded with the word "Table". (For example, Table 1-5 would designate the fifth table in the first section.)

3.2.2.4.7.10 Charts and Graphs. Material that includes charts, graphs, curves, or other pictorial matter shall be treated as a chart but shall be considered and processed as an illustration. Numbering shall be the same as for figures and shall be included in the list of illustrations.

3.2.2.5 Page Content Headings, Technical Manuals.

3.2.2.5.1 Location and Layout. Each basic reproduction page shall have page content headings consisting of paragraph numbers. The first paragraph that begins on a left-hand page shall be shown by number in the upper outside corner of the page and shall be set flush left. The last paragraph that begins on a right-hand page shall be shown by number in the upper outside corner of the page and shall be set flush right.

3.2.2.5.2 Where two or more sections appear on one page, the number of the first and last paragraphs any portion of which appear on that page shall be shown. (For example: 2-1 to 3-4.)

3.2.2.5.3 When a page contains only one section complete, the first paragraph number shall be the sole identification.

3.2.2.5.4 When one or more illustrations occupy the entire page, the page content heading shall be assigned as follows:

a. If the page is a right-hand page, the last paragraph number of the preceding left-hand text page shall be used.

b. If the page is a left-hand page, the first paragraph number of the following right-hand text page shall be used.

3.2.2.5.5 Each page of an index shall show in the upper outside corner the first and last main items listed on the page.

3.2.2.5.6 Section Titles and Headings. Section titles shall be centered horizontally on the page. The section titles shall conform to the requirements of the applicable technical content specification. Each section shall begin on a new page, except that when a section is less than a page in length, the succeeding section may begin on the same page.

3.3 Illustrations.

3.3.1 General Requirements. Illustrations shall be used to clarify text, to present phases difficult to describe by text alone, to call attention to details, and to furnish pictorial identification of parts and tools required, using the minimum number essential for such purposes. Illustrations shall be planned for publication to as small a scale as will satisfactorily portray the required details. They shall not be signed by an artist. Illustrations serving no instructional function shall not be used. All illustrations shall be referenced in the text. When reference is to a multiple sheet illustration, the figure number and sheet number shall be specified.

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3.3.1.1 Location. Illustrations shall be located as near as possible to related text, except where this would require unnecessary duplication of illustrations.

3.3.1.2 Photographs. Photographic illustrations shall faithfully reproduce all details, be continuous tone (unscreened), and clearly show the subject matter. They shall be retouched to the least degree consistent with obtaining illustrations which will clearly convey the intended information and print well by the offset method.

3.3.1.3 Exploded Views. Line illustrations of exploded views are preferred to photographic illustrations of exploded views since they normally give better results in showing small intricate parts. However, if photographic or other methods are used small intricate parts shall be clearly defined.

3.3.1.4 Charts and Graphs. The use of charts and graphs to present graphically or pictorially rates of flow, lubricating points, power curves, etc., is desirable.

3.3.1.5 Cartoons. Cartoons (exaggerated or comic figure drawings) shall be used in manuals only to emphasize particularly hazardous points and only when authorized by the Procuring Activity. They shall not be assigned figure numbers.

3.3.1.6 Fold-Over Pages. Fold-over pages shall be used only for diagrams, charts,or general arrangements for major systems such as a missile launching and handling system which cannot be reduced for satisfactory presentation on single pages. Before planning the use of fold-over pages, the layout of the material shall be considered to determine whether improved presentation would result if it were simplified or divided into more usable units for reproduction in less extended space.

3.3.1.7 Color.

3.3.1.7.1 General Use. Color shall be used only where absolutely necessary to clarify functional operations. Cross-hatching, similar drafting methods, or other shading media shall be employed in lieu of color unless the complexity of the illustration demands the use of color. Under no circumstances shall color be used for backgrounds or other decorative purposes. A legend on the illustration shall explain the significance of each color used. Color shall not be used in cartoons, notes, cautions, and warnings.

3.3.1.7.2 Selection of Colors. When color is used, its selection shall be based on the primary colors (red, blue, and yellow) by which required color values are normally obtained in printing. Screens shall be used whenever practicable to reduce the number of printing impressions required. For example, if red is used, a 50-percent screen of the red would produce a tint which would serve in lieu of an additional color. This screen shall be produced by adding white dot (Zip-a-tone, Ben Day, etc.) screen, on original color separation overlay. The use of colors in combination may be employed. For example, combining yellow and blue if used for two of the colors, to form a third color, green; or red and yellow to obtain orange; red and blue to obtain purple, etc. When colors are to be combined, the following information must accompany the reproducible copy: ink type, ink number, and the order in which the colors shall be printed. Color swatches shall be furnished if exact shading is necessary or desirable.

3.3.1.7.3 Preparation of Color Overlays. Area to be colored in the final reproduction shall be prepared in permanent black or red on overlays of clear or frosted (matte-finish) acetate sheeting. Areas to appear in color shall be pre-

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pared for each basic color. Tints of a color shall be prepared on the basic color overlay as described in 3.3.1.7.2. Overlays shall be carefully registered to the basic or key illustration. (See figure 14.) Register marks shall be centered on at least three sides as close as practicable to the image area. If exact shading is pertinent, color swatches, ink type, ink number, and order of printing shall be furnished.

3.3.1.7.4 Preparation of Color Overlay Negatives. When approved by the Procuring Activity, color overlay separation negatives may be submitted in lieu of overlays. When this method is used, the following requirements are applicable:

3.3.1.7.4.1 A basic or key illustration shall show the illustration appropriately colored as a guide to show how it should appear in final printed form. Color swatches, ink type, ink number, and order of printing shall be furnished if exact shading is pertinent.

3.3.1.7.4.2 Accompanying the key illustration, there shall be a separate negative for each color used, plus one negative for black. Each negative shall be opaqued as necessary to blank out all except the pertinent color. For example, if red is one of the colors, the reproduction area in the appropriate negative must be completely opaqued except for the portions to be colored in red in the final printed publication.

3.3.1.7.4.3 Each key illustration, together with its accompanying set of negatives, shall be in its own envelope. The envelope, the key illustration, and each negative shall be marked as to manual identifying number, figure number, page number, and security classification (if any). Register marks shall be centered on at least three sides of the key illustration as close as practicable to the image area.

3.3.1.7.5 Color Coding Reference. Reference to color coding of chassis wiring shall be in accordance with Standard MIL-STD-122.

3.3.1.7.6 Legends. Artwork containing color shall contain a legend as part of the illustration. (See 3.3.1.7.1.) The legend shall contain information to identify usage of the color in the illustration.

3.3.1.7.7 Color Negative Identification. Color separation negatives shall contain the following information placed outside of the reproduction area: Manual number, figure number, page number, and name of color involved.

3.3.1.8 Unacceptable Illustration Copy. The following types of illustration copy will not be accepted: Copy prints made from negatives containing a halftone screen; continuous tone film negatives; film positives (except for color overlay preparation, or for the black key drawing of a color illustration); combination line and halftone illustrations, except as permitted by 3.3.2.2.8; brown-line prints, photostats, bromides, and prints made by similar reproduction processes. Continuous tone illustrations intended for outline-halftone (silhouette) printing are unacceptable, except where the illustrations cannot be prepared for printing as a square-finish halftone. Bleed illustrations and line illustrations containing weak or broken lines are not acceptable.

3.3.2 Manner of Preparation.

3.3.2.1 Completeness. A complete set of artwork shall be submitted for each basic manual. When a basic continous tone illustration is used exactly in the same size in more than one place in a manual only one piece of artwork must be furnished with necessary identifying information. When a basic continuous tone illustration is to be used for two or more figures with different overlays of lead in

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arrows or text, a separate basic continuous tone print with its appropriate overlay shall be furnished for each applicable figure. Where original art is furnished, the overlays, maximum of three, may be attached respectively, at the top left, and right sides of the original art. Illustrations shall be furnished with all mechanical indications required for accurate offset reproduction.

3.3.2.2 Continuous Tone Illustrations (Halftone). The exact location and final size of each continuous tone illustration (halftone) shall be indicated on the reproducible copy by hairline rules or fine lines drawn in black ink locating the top and bottom limits of the illustration or by black or red masks mounted in the reproducible page.

3.3.2.2.1 Submission of Illustrations for Halftone Reproduction. Illustrations intended for halftone reproduction shall be submitted as continuous tone (unscreened) photocopy prints or original art. The subject to be depicted shall preferably be 1-1/2 times final reproduction (printing) size. As many as possible of the illustrations shall be scaled for uniform percentages of reduction, to save time and expense in making negatives.

3.3.2.2.2 Mounting. All continuous tone (halftone) illustrations and reproducible pages containing color illustrations with more than one color overlay shall be mounted on suitable mounting boards not larger than 14 by 17 inches, except the art for fold-over pages requiring larger mountings. They shall be protected by an inner tissue and an outer heavy paper cover securely affixed to the mount along the top edge to provide a protective covering or flap. Tissue stock with an oily or wax content shall not be used for the inner overlay. The distance between any border of the illustration proper and the corresponding edge of the mount shall normally be 1-1/2 inches on each side and across the top and 2 inches on the bottom. In no case shall the distance between the edge of the artwork and the edge of the mount be less than 1 inch. Edges of the artwork shall be parallel to the edge of the mounting board. Each figure shall be identified in accordance with 3.3.2.2.4. (See figure 14.)

3.3.2.2.3 Artwork shall be indicated for cropping by use of crop marks placed on the marginal area as close as practicable to the image area of the illustration to indicate the portion of the figure to be included in the final reproduction. (See figure 14.) The proportion shall agree with the proportion indicated on the reproducible copy. The horizontal reproduction size shall be indicated in picas between crop marks. The word "Top" shall be placed on the mounting or margin to indicate correct position of the illustration.

3.3.2.2.4 Identification of Mounted Continuous Tone (Halftone) Illustrations. The figure number, page number, security classification (if any), and manual identifying number shall be indicated in the upper right-hand corner of the outer page cover and on the face of the mount, as close as practicable to the illustration and preferably not more than 1/16-inch outside the bottom crop line of the illustration.

3.3.2.2.5 If an illustration occupies an entire page, the page shall be prepared with complete mechanical indications, such as security classification, manual number, page content heading, illustration size and location indications, figure number and title, page number, and any other text matter. Artwork for the halftone illustration shall be submitted separately.

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3.3.2.2.6 Continuous Tone (Halftone) Illustrations in Changes. When a page of reproducible copy containing a halftone illustration is furnished as part of a change or revision and the halftone illustration is not affected by the change or revision, no new art need be furnished. When the contract requires submittal of offset negatives, the page negative shall have the screened halftone negative mortised in place.

3.3.2.2.7 Combination Artwork Illustrating Details of Equipment. Where it is necessary to show details of equipment with callouts, lines, arrows, etc., such art shall be prepared for reproduction and printing by photographing through a halftone screen. The artwork shall be furnished as either original artwork or good clear continuous tone print with the callouts, lines, arrows, etc. either affixed directly on the art or placed on a clear acetate overlay registered with the art.

3.3.2.2.8 Combination Artwork Illustrating Steps and Procedure. Combination artwork that illustrates steps of procedure may be submitted as line artwork mounted on the reproducible page. When such artwork is submitted, it shall be produced by making a continuous tone negative from the original continuous tone artwork. This negative shall be reduced to the correct size for the space allotted on the reproducible page. Then, the continuous tone negative shall be used to make a contact print through the use of a magenta screen. All prints shall be clean, clear, and sharp in dot structure with no blurs, fill-ins, or distortion in dot structure.

3.3.2.2.8.1 When reproducible copy is prepared at reproduction size or at 10-percent oversize, a 120-line screen shall be used to convert the tone negative to a line print, so that the final printed illustration will approximate a 120- or a 140-line screen pattern, respectively. When reproducible copy is prepared at 20- or 25-percent oversize, a 100-line screen shall be used to convert the tone negative to a line print, so that the final printed illustration will approximate a 128- or a 139-line screen pattern, respectively.

3.3.2.2.8.2 When tone artwork is converted to line, the properly identified continuous tone artwork shall also be delivered for each piece of tone artwork converted to line.

3.3.2.2.8.3 The use of a magenta screen to convert tone artwork to line for illustrating steps of procedure shall not be contrued as permission to convert any other tone artwork in the manual to line or to prepare elaborate layouts of silhouetted tone artwork.

3.3.2.2.8.4 Related text may be presented as a part of an illustration only in illustrated steps of procedure wherein such procedural steps are shown in a series of tone, combination, or line drawings which, as a group, make up a single illustration. In such cases text shall be properly spaced on the basic reproduction page. It shall not appear on the separately mounted art.

3.3.2.3 Line Illustrations. Line illustrations shall have sharp, clean black lines on a white background. Lines shall be strong and unbroken and shall have sufficient weight to insure good reproduction.

3.3.2.3.1 Submission of Line Illustrations. Line illustrations shall be furnished as unscreened copy prints or original art, in final reproduction (printing size) except that prints for illustrations of oversize typewritten reproducible pages shall be scaled to fit the oversize dimensions, and shall be dry mounted or mortised in proper spaces on the repro-

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ducible page.

3.3.2.3.2 Diagrams. Lettering on diagrams shall be produced by machine composition or with the aid of a mechanical device. Successive lines of lettering shall be separated by not less than two-thirds the height of the letters. Parallel lines on wiring and schematic diagrams shall be separated by not less than 1/16 inch when printed.

3.3.2.3.3 Reproducible text pages shall not be mounted except when the mounting of line art on a sheet tends to wrinkle or otherwise distort the reproducible copy. However, a reproducible text page shall be mounted if there is more than one overlay for the page.

3.3.2.3.4 Index Numbers, Reference Designators, and Nomenclature on Illustrations.

3.3.2.3.4.1 Use for Manuals. Significant features or components of illustrations in manuals shall be identified by index references (this term includes index numbers and reference designations) or brief nomenclature. (See figures 15 and 16.) Nomenclature and word forms shall be consistent with terms used in the text.

3.3.2.3.4.2 Size and Legibility. Index references and lettering shall be neat and clearly legible, and shall be prepared with the aid of a mechanical device, or shall be typeset characters. They shall be planned to reproduce in a size between 8- and 10-point type (approximately 1/8-inch high) and shall be consistent within the manual. On schematic diagrams, 6-point type is permissible provided the characters are legible.

3.3.2.3.4.3 Means of Application. Index references, nomenclature, lines and arrows may be drawn or placed directly on the illustration or on clear acetate overlays. If they are placed directly on a halftone illustration, the illustration must be exceptionally well retouched for clarity; otherwise they must be placed on overlays. To assure clear definition when passing through light and dark areas, lines and arrows shall be drawn in black with at least one edge outlined in white. Lines shall not be placed parallel to principal surfaces, edges of illustrated objects, edges of illustrations, or to lines of zip-a-tone patterns. Lines must be straight; curves, doglegs, or bends are not permitted.

3.3.2.3.4.4 Lines and Arrows. A line (and an arrowhead when necessary to clearly indicate the detail concerned) shall lead from the index reference or nomenclature to the item to which it pertains.

3.3.2.3.4.5 Index Number Sequence. Where index numbers are used, each illustration shall be handled independently, with index numbers assigned consecutively starting with Arabic numeral 1. Index numbers shall be in numerical sequence in any list referenced to the illustration. Whenever practicable, index numbers shall also be in sequence on the illustration, and arranged in a vertical, horizontal, or diagonal line instead of being scattered. In parts breakdowns, when index-numbered items are added by changes, letter-suffixed numbers shall be used. (Example: Items added between index numbers 10 and 11 should be given index numbers 10a through 10z, then 10aa, 10ab, etc.) Letters and numbers shall not be superimposed upon lines.

3.3.2.3.4.6 Line Illustrations in Changes. When a page containing a line illustration is furnished as a part of a change or revision and the line illustration is not affected by the change or revision, a photocopy print of the illustration as previously furnished shall be assembled on the reproducible page.

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3.3.2.4 Two-Page Spread. When an illustration or table requires more than the space available on one page and is not within the requirements of 3.3.1.6, it shall be prepared as a two-page spread.

3.3.2.5 Make-Up of Fold-Over Pages. When fold-over pages are used, they shall be made up as indicated in figures 12 and 13. They shall be held to a two-page fold-over where practicable. They shall not exceed a maximum image area including marginal copy of 55 picas high and 234 picas (39 inches) long. When a fold-over is planned for printing with a blank apron (normally used only for schematic diagrams) the absolute maximum image area including marginal copy shall not exceed 55 picas in height and 180 picas (30 inches) in length. When an illustration is of such size that it cannot be legibly reproduced within the maximum image area, it should be rearranged for reproduction on two or more pages. Sufficient overlap of the illustration shall be included on each of the pages to insure continuity; breaks in lines shall be adequately identified, and the breaking points shall be indicated. Fold-over pages shall be backed up by a diagram or chart similar in size to that on the front, except that those containing schematic diagrams normally will not be backed up.

3.3.2.5.1 Fold-Over Pages Requiring Horizontal and Vertical Folds. Fold-over pages requiring horizontal (fold-up) and vertical folding shall not be prepared except as specifically approved by the Procuring Activity. When approved, the illustration area may be made higher that 55 picas, but in no case shall the image area including marginal copy exceed 115 picas (19-1/6 inches) high by 234 picas (39 inches) in length, without a blank apron. When a blank apron is planned, length shall not exceed 180 picas (30 inches).

3.4 Reproducible Copy Requirements.

3.4.1 Kind of Reproducible Copy. Unless otherwise specified a complete set of reproducible copy complete in all aspects for offset printing shall be prepared.

3.4.2 Basic Reproducible Page.

3.4.2.1 The basic reproducible page shall be the sheet containing the complete text, line illustrations, and marginal copy for a manual page.

3.4.2.2 The reproducible page shall be of white paper stock imprinted in the type medium selected with the necessary space allotted for illustrations.

3.4.2.3 Shading media shall not be used to outline text or headings. Tint blocks shall not be used in reproducible copy solely for aesthetic design or decorative devices.

3.4.3 Typography. The instruction and marginal copy shall be typeset or produced on office composing machines (cold-type). The two may be combined only to the extent that when the text portion is produced on office composing machines, headings and other requirements for display faces may be typeset. This does not exclude use of office composing machines within illustrations for callouts, index numbers, etc. The type sizes and faces shall be in accordance with figure 1, sheets 1 through 6.

3.4.3.1 Leading and Spacing. Layout shall conserve space without lessening usability or clarity of material. Blank pages and spaces in manuals shall be avoided wherever possible. Leading and spacing as required shall be used for best readability and conservation of space. However, double spacing of text within a paragraph or subparagraph and similar

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extravagance shall not be permitted. Slight variations will naturally occur during layout in order to avoid poor layout practices such as:

(a) Paragraphs divided so that the first line is alone at the bottom of a column or the last line begins on a new page.

(b) Subheads located so that they fall on the last line of a page or column.

(c) Notes, cautions, and warnings divided so that first lines appear in one column and remaining lines in another.

3.4.4 Reproduction Area. Maximum image area for single pages shall be 40 picas wide by 55 picas deep. Material prepared on office composing machines (cold type) shall be prepared to a scale which will result in a 40-pica by 55-pica image area when reduced photographically for offset printing.

3.4.5 Instruction Preparation.

3.4.5.1 Columnar Arrangement. The text shall be in two columns. Each column shall be 19 picas (3-1/8 inches) wide with a 2-pica space (3/8 inch) between columns. Right-hand margins shall be justified, except as specified in 3.1.1.1.

3.4.5.2 Indentations. There shall be no indentations within columns except that steps of procedure shall be indented 1 em in the first line only and notes, cautions, and warnings shall be indented 2 ems on each side. Paragraph numbers preceding main paragraph headings shall be flush with the left vertical margin of the column. Headings of more than one line shall have all additional lines flush on the left with the beginning of the heading.

3.4.5.3 Section on Right-Hand Page. Sections shall begin on a right-hand page. If a full page illustration is required at the beginning of a section, it shall be placed on the second or succeeding pages of the section except as specified in 3.2.2.4.7.4.

3.4.5.4 Section Titles and Headings. Section titles shall be centered horizontally on the page.

3.4.6 Marginal Copy.

3.4.6.1 Technical Manual Number. Each basic reproducible page shall bear the manual identifying number assigned by the Procuring Activity. The identifying number shall be centered at the top of the reproducible page.

3.4.6.2 Page Content Heading. Each basic reproduction page shall have content headings in accordance with 3.2.2.5.

3.4.6.3 Arrangement of Marginal Copy. The page content headings shall be at the top of the page, flush with the outside reproduction margin. The manual identifying number shall be centered at top of the page within the reproduction area. The page number shall be at the bottom outside corner of the page, flush with the vertical margin of the text or illustration, and within the reproduction area. The security classification markings shall be at the top and bottom of the page, centered above and below the text but within reproduction area. To permit this, the identifying number shall be placed below the security classification marking.

3.4.7 Requirements for Manuals of 8 Pages or Less. Manuals of 8 pages or less shall:

(a) Not have table of contents, list of effective pages, or list of illustrations.

(b) Begin sections on left or right-hand pages with no blank pages (more than one section on a page when possible).

(c) Have pages, illustrations, and paragraphs numbered consecutively with single Arabic numeral.

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(d) Begin text below abbreviated title panel.
(e) Be revised in lieu of changed.

3.4.8 Publication Date. The publication date shall be assigned in accordance with Specification MIL-H-7863(Aer). The publication date shall not appear on text or illustration pages of a basic manual or revision.

3.4.9 Changed Pages.

3.4.9.1 Change Date. Changed pages shall carry in the lower left-hand corner of odd numbered pages or the right-hand corner of even numbered pages, the change date which shall conform to the change date on the title page. This date in approved military form shall be on the same line as the page number. (See figure 17.)

3.4.9.2 Change Symbols. Changed text, including legends, figure titles, and deletion notes, shall be indicated by a 1/16-inch black vertical line in the outer margins (the left margin for left-hand columns and the right margin for right-hand columns) approximately 1/16 inch from the text. (See figure 17.) Symbols are to indicate only changes made in the current change. The change symbol shall not be used for illustration revisions, minor typographical corrections, changes in physical location of material, changes in running heads, blank space resulting from shortened copy, or changes in alphabetical indexes.

3.4.9.3 Change To Add Pages. If additional copy is added to a right-hand page, the overrun shall be carried over to the next left-hand page, and the overrun from this page shall be placed on an added page.

3.4.9.3.1 Changed or added pages shall be prepared in the same style and format as the latest issue of the manual concerned.

3.4.9.3.2 Unchanged pages including those used only for "back-up" shall not be resubmitted.

3.4.9.3.3 Changed or added pages shall reflect the current security classification of the information contained therein at the time the change is prepared.

3.4.10 Revisions. Revisions (see 3.1.3.3) shall be prepared in the same manner as basic manuals except as specified herein. All previously modified page, paragraph, and figure numbers shall be eliminated in a revision. All previous change symbols shall be eliminated. Change symbols shall be used in a revision for new or altered material involving only such critical items as safety of flight, correction of significant errors in text that jeopardize personnel or equipment, and production changes. Major replacement of material such as whole chapters or sections, or major sections of chapters, shall not be indicated by change symbols in the text. However, a change symbol shall indicate the insertion of new or revised material in the table of contents, list of illustrations, and list of tables in the front matter.

3.4.10.1 Unchanged reproducible pages shall not be submitted as a part of a revision unless specifically required in the contract except as provided in 3.4.10.2. Unchanged pages required for printing the revision should be available from the master reproducible files for the manual and should be obtained from this file.

3.4.10.2 When a revision is being prepared and the security classification has been changed since the previous edition, such change in classification shall be reflected in the revision with the page markings as required by 3.1.5.2 and all pages required for such revision shall be submitted.

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3.4.11 Supplements. Supplements shall be prepared in the same style and format as the manuals they supplement. They shall be treated as complete and separate manuals with respect to page, paragraph, and figure numbering and other format requirements set forth in the preceding paragraphs.

3.4.11.1 Supplements shall consist of sections identical in title to the sections comprising the manuals supplemented. Where supplementary data for any section are not required, that section of the supplement shall refer to the supplemented manual with a statement that there are no supplementary data.

3.4.11.2 Supplements shall contain an introductory statement explaining the nature of the supplement and the relationship of the supplement to the manuals supplemented. Reference shall also be made to the identifying number and title of the manual(s) supplemented.

3.4.11.3 Reference to the supplemented manual(s) shall be made as appropriate throughout. Such reference shall use the identifying number, title, and, when appropriate, the section number of the supplemented manual. Reference shall not be made to specified page, paragraph, or figure numbers.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements. Government inspection and approval of any portion of the material during the course of preparation shall not be construed as acceptance of the completed product.

4.2 Material furnished in accordance with this specification shall be inspected by the supplier for conformance to the applicable requirements of the governing documents.

4.2.1 Information and instructions, both textual and illustrative, originated and furnished by the supplier shall be reviewed and approved by the supplier for technical and engineering accuracy and adequacy, and for conformance to the requirements of the technical content specification. Evidence of such review and approval shall be made available to the Government inspector, upon request, at the time the material furnished hereunder is presented for Government acceptance.

4.3 Material furnished in accordance with this specification shall be subject to inspection, verification, and approval or disapproval by the Government prior to acceptance.

5. PREPARATION FOR DELIVERY

5.1 Reproducible copy shall be packed unfolded in commercial shipping containers that will meet the requirements of the Consolidated Freight Classification Rules in effect at the time of shipment. Reproducible copy and artwork shall be arranged in groups according to size and so packaged that all material will lay flat and withstand rough handling. Fillers or thicknesses of corrugated board shall be added to insure stability.

5.2 Classified material shall be packed for delivery in accordance with the requirements of DD-441 (Attachment).

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5.3 Reproducible copy delivered to Navy shall be accompanied by Form NAVAER 2712, "Reproducible Manuscript/Receipt/Acceptance" and Form NAVAER 2712A, "Manuscript Checkoff List".

5.4 Each shipping container containing reproducible copy shall have affixed thereto a copy of the applicable letter of transmittal and assembly sheet. The words "Reproducible Copy (Manuscripts)" and the following information shall appear on the outside of each shipping container:

Addressee
Number of containers in shipment
Technical manual number
Number of applicable specification
Contract or order number
Name of contractor

6. NOTES

6.1 Intended Use. The intended use of this specification is to cover the general requirements for preparation of technical manuals used by the Bureau of Naval Weapons.

6.2 Contractors shall always retain a duplicate copy (negatives, photocopy, copy prints, etc.) of all material furnished for the life of the contract. This is to facilitate replacement in case of loss or damage in shipment and for use in preparing changes.

6.3 Definitions. The following definitions apply to the terms as they are used in this specification:

6.3.1 Back-Up Page. A back-up page is the second or reverse side of a printed page.

6.3.2 Basic Manual. A basic manual is the initial edition of a technical manual.

6.3.3 Change. (Formerly BuAer Revision.) A change is any alteration in a portion of a manual already in existence but which is not large enough to require issuance of a revision.

6.3.4 Change Back-Up Page. A change back-up page is that page which must be printed to complete both sides of a leaf of a manual when one side has been altered and the other side is unchanged.

6.3.5 Chapter. A chapter is the primary division within a technical manual, when required by the technical content specification, and may consist of one or more sections.

6.3.6 Fold-Over (Fold-Out, Fold-In). A fold-over page is a page used to present information which must be larger than the trim size of the manual and therefore must be folded to the size of the single page for incorporation into the manual. Sometimes called "gate-fold" and "insert".

6.3.7 Leaf. A leaf is a sheet, each side of which is a page.

6.3.8 Manuscript Copy. Manuscript copy is typewritten copy and illustrative material, not intended for direct reproduction, furnished to the Procuring Activity for review and any necessary changes prior to preparation of reproducible copy.

6.3.9 Marginal Copy. Marginal copy is identifying information other than text or illustrations appearing on a page.

6.3.10 Original Art. Original art is the original drawing or photographic print retouched as necessary to insure first class reproduction, or a photographic print of the original drawing or retouched photograph which retains all the qualities of the original.

6.3.11 Page. A page is one side of a leaf of a manual.

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6.3.12 Photocopy Print. A photocopy print is a photographic copy of line drawings, artwork, or photographs used for reproduction copy.

6.3.13 Procuring Activity. The Procuring Activity is the office within the Department of Defense delegated the authority to issue a contract.

6.3.14 Revision (Formerly BuAer Reissue). A revision is a second or subsequent edition of a technical manual which supersedes the preceding edition.

6.3.15 Section. A section is the primary division within a technical manual when required by the technical content specification. A section may be a subdivision within a chapter. (See 6.3.5.)

6.3.16 Supplement. A supplement is a subsidiary manual which complements information in a technical manual.

6.3.17 Technical Manual. A technical manual is a publication containing a description of equipment or a weapon with instructions for effective use including one or more of the following sections as required:

(a) Instructions covering initial preparation for use
(b) Operating Instructions
(c) Maintenance Instructions
(d) Overhaul Instructions
(e) Parts List or Illustrated Parts Breakdown
(f) Related technical information or procedures exclusive of those procedures of an administrative nature.

6.3.18 Two-Page Spread. A two-page spread is matter which spreads across and fills the entire two-page opening of a manual causing the two pages to appear as a single unit. (See 3.3.2.4.)

6.3.19 Unscreened Print. An unscreened print is a continuous tone photograph intended for reproduction as a halftone.

NOTICE. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing activity:
Navy-Wep

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Copy	Typeset 1/	DSJ Varityper	IBM, Varityper or Typewriter 2/
Title page	Reference MIL-H-7863(Aer)	Reference MIL-H-7863(Aer)	Reference MIL-H-7863(Aer)
A page	Reference MIL-H-7863(Aer)	Reference MIL-H-7863(Aer)	Reference MIL-H-7863(Aer)
Flyleaf			
Center head	14 pt Vogue Bold caps	670-12 caps	text type caps 3/
Body	text type	text type	text type U&lc
Foreword			
Center head	14 pt Vogue Bold caps	670-12 caps	text type caps 3/
Body	text type	text type	text type U&lc
Contents, illustrations, tables, safety summary			
Center head	14 pt Vogue Bold caps	670-12 caps	text type caps 3/
Body	text type	text type	text type U&lc
Text, body	10 pt on 12 Modern flush left, except for steps of procedure which indent 1 em	600-10 flush left, except for steps of procedure which indent 1 em	2/flush left, except for steps of procedure which indent 1 em

1/Acceptable substitute type faces are:
Vogue Bold - Futura, Spartan, Twentieth Century, Sans Serif, or similar
Modern - Garamond, Bodoni, Bookman, or similar
DSJ 670 or 660
DSJ 780 or 680
DSJ 600 or 620
DSJ 605 or 625
} or similar

2/As stated in par. 3.2.2.1.2 acceptable type faces are:
IBM Electromatic
Bold Face - 25% oversize
Bold Face No. 2 - reproduction size
Modern - 10% oversize
Varityper
250-10-12 - 25% oversize
Typewriter
Pica - 25% oversize
Elite - 20% oversize

3/Typeset or art type may be used for these items, par. 3.4.3.

Figure 1 (Sheet 1). Type Faces, Sizes, and Details of Preparation. (Par. 3.2.2.1.1.1)

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Copy	Typeset 1/	DSJ Varityper	IBM, Varityper or Typewriter 2/
Text, emphatic	10 pt bold of text type	780-10	text type underscored
First order head	12 pt Vogue Bold caps, on a line alone	670-10 caps, on a line alone	text type caps, 3/ on a line alone
Second order head	10 pt Bold Face of text type U&lc, run-in with text, followed by a period	780-10 caps, run-in with text, followed by a period	text type caps, run-in with text, followed by a period
Third order head	10 pt of text type caps and sm. caps U&lc, run-in with text, followed by a period	600-10 caps, run-in with text, followed by a period	text type U&lc, underscored, run-in with text, followed by a period
Fourth order head	10 pt italic of text type U&lc run-in with text, followed by a period	605-10 U&lc run-in with text, followed by a period	text type U&lc run-in with text, followed by a period
Equations and fire control symbols	10 pt italic of text type U&lc	605-10 U&lc or 710-10	text type U&lc
Footnotes	6 pt text type U&lc	600-7 U&lc	text type U&lc

1/ Acceptable substitute type faces are:
Vogue Bold - Futura, Spartan, Twentieth Century, Sans Serif, or similar
Modern - Garamond, Bodoni, Bookman, or similar
DSJ 670 or 660
DSJ 780 or 680
DSJ 600 or 620
DSJ 605 or 625
} or similar

2/ As stated in par. 3.2.2.1.2 acceptable type faces are:
IBM Electromatic
Bold Face - 25% oversize
Bold Face No. 2 - reproduction size
Modern - 10% oversize
Varityper
250-10-12 - 25% oversize
Typewriter
Pica - 25% oversize
Elite - 20% oversize

3/ Typeset or art type may be used for these items, par. 3.4.3.

Figure 1 (Sheet 2). Type Faces, Sizes, and Details of Preparation. (Par. 3.2.2.1.1.1)

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Copy	Typeset 1/	DSJ Varityper	IBM, Varityper or Typewriter 2/
Table number and title	10 pt Bold Face of text type U&lc, centered pyramid style	780-10 U&lc, centered pyramid style	text type caps, centered pyramid style
Table subheads and body	8 pt text type U&lc. To conserve space, subheads may be set in smaller size, but no smaller than 6 pt.	600-8 U&lc. To conserve space, subheads may be set in smaller size, but no smaller than 6 pt.	text type U&lc
Warning			
Center head	12 pt Vogue Bold caps	670-12 caps	text type caps 3/ 4/ underlined
Body	text type, indented 2 ems on each side	text type U&lc, indented 2 ems on each side	text type U&lc indented 2 ems on each side
Caution			
Center head	10 pt Bold Face of text type caps	780-10 caps	text type caps 3/ 4/
Body	text type indented 2 ems on each side	text type U&lc, indented 2 ems on each side	text type U&lc indented 2 ems on each side

1/ Acceptable substitute type faces are:
Vogue Bold - Futura, Spartan, Twentieth Century, Sans Serif, or similar
Modern - Garamond, Bodoni, Bookman, or similar
DSJ 670 or 660
DSJ 780 or 680
DSJ 600 or 620
DSJ 605 or 625
} or similar

2/ As stated in par. 3.2.2.1.2 acceptable type faces are:
IBM Electromatic
Bold Face - 25% oversize
Bold Face No. 2 - reproduction size
Modern - 10% oversize
Varityper
250-10-12 - 25% oversize
Typewriter
Pica - 25% oversize
Elite - 20% oversize

3/ Typeset or art type may be used for these items, par. 3.4.3.

4/ See figure 2 for alternate method for presentation of display heads for CAUTION and WARNING.

Figure 1 (Sheet 3). Type Faces, Sizes, and Details of Preparation. (Par. 3.2.2.1.1.1)

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Copy	Typeset 1/	DSJ Varityper	IBM, Varityper or Typewriter 2/
Note			
Head	10 pt text type caps, run-in with text, followed by colon	10 pt text type caps, run-in with text, followed by colon	text type caps, run-in with text, followed by colon
Body	10 pt text type U&lc indented 2 ems on each side	10 pt text type U&lc indented 2 ems on each side	text type U&lc indented 2 ems on each side
Chapter or section numbers and titles, appendix numbers and titles	14 pt Vogue Bold caps, centered	670-12 caps, centered	text type caps, 3/ centered
Section number and title within a chapter	12 pt Vogue Bold U&lc centered	670-12 U&lc, centered	text type U&lc, centered
Index			
Center head	14 pt Vogue Bold caps	670-12 caps	text type caps 3/
Body	8 pt text type U&lc	8 pt text type U&lc	text type U&lc
Figure captions	10 pt Vogue Bold Italics U&lc, centered pyramid style	605-10 U&lc, centered pyramid style	text type U&lc, centered pyramid style

1/ Acceptable substitute type faces are:
Vogue Bold - Futura, Spartan, Twentieth Century, Sans Serif, or similar
Modern - Garamond, Bodoni, Bookman, or similar
DSJ 670 or 660
DSJ 780 or 680
DSJ 600 or 620 } or similar
DSJ 605 or 625

2/ As stated in par. 3.2.2.1.2 acceptable type faces are:
IBM Electromatic
Bold Face - 25% oversize
Bold Face No. 2 - reproduction size
Modern - 10% oversize
Varityper
250-10-12 - 25% oversize
Typewriter
Pica - 25% oversize
Elite - 20% oversize

3/ Typeset or art type may be used for these items, par. 3.4.3.

Figure 1 (Sheet 4). Type Faces, Sizes, and Details of Preparation. (Par. 3.2.2.1.1.1)

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Copy	Typeset 1/	DSJ Varityper	IBM, Varityper or Typewriter 2/
Running heads	12 pt Vogue Bold caps, flush right on odd pages, flush left on even pages	670-12 caps, flush right on odd pages, flush left on even pages	text type caps, flush right on odd pages, flush left on even pages 3/
Fixed heads	12 pt Vogue Bold caps, centered at top and bottom of page	670-12 caps, centered at top and bottom of page	text type caps, centered at top and bottom of page 3/
Page numbers	12 pt Vogue Bold caps, flush right on odd pages, flush left on even pages	670-12 caps, flush right on odd pages, flush left on even pages	text type caps, flush right on odd pages, flush left on even pages 3/
Change date	12 pt Vogue Bold caps, flush right on odd pages, flush left on even pages	670-12 caps, flush right on odd pages, flush left on even pages	text type caps, flush right on odd pages, flush left on even pages 3/

1/ Acceptable substitute type faces are:
Vogue Bold - Futura, Spartan, Twentieth Century, Sans Serif, or similar
Modern - Garamond, Bodoni, Bookman, or similar
DSJ 670 or 660
DSJ 780 or 680
DSJ 600 or 620
DSJ 605 or 625
} or similar

2/ As stated in par. 3.2.2.1.2 acceptable type faces are:
IBM Electromatic
Bold Face - 25% oversize
Bold Face No. 2 - reproduction size
Modern - 10% oversize
Varityper
250-10-12 - 25% oversize
Typewriter
Pica - 25% oversize
Elite - 20% oversize

3/ Typeset or art type may be used for these items, par. 3.4.3.

Figure 1 (Sheet 5). Type Faces, Sizes, and Details of Preparation. (Par. 3.2.2.1.1.1)

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Spacing Between	Cold Type Copy (lines extra)	Typeset Copy (picas)
Running head and text	1	1
Running head and Section number	6	6
Running head and heads for contents, illustrations, and so forth	6	6
Section number and title	1	1
Section title and text	2	2
Text and folio	1	1

Spacing Between	Cold Type Copy (lines extra)	Typeset Copy (picas)
Text and figure caption or table title	1	1
End of paragraph and text paragraph head	1	1
Between paragraphs	1	1
Section numbers and titles within a chapter and preceding and following material	1	1
First order head and following paragraph	1	1
End of paragraph and first order head	2	2

NOTE: These specifications are for copy that will reproduce same size. When oversize pages are used, the reduced page should approximate these spacing requirements.

Spacing requirements may be relaxed to avoid widow lines and other unacceptable layouts.

Figure 1 (Sheet 6). Type Faces, Sizes, and Details of Preparation. (Par. 3.2.2.1.1.1)



Secure air supply to system and bleed pressure by extending and retracting snubbers to prevent uncontrolled movement from crushing operator. Attach appropriate warning tag.



Note exact order in which parts are removed. Cushioning feature described in chapter 2 depends on correct assembly.

Figure 2 Alternate Method for CAUTION and WARNING Display Heads. (Par. 3.1.14)

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CHAPTER 2
DESCRIPTION

Section 2-1. Launcher Mk 112 Mod 0

2-1. (C) Launcher Mk 112 Mod 0 consists of Stand Mk 107 Mod 0, Carriage Mk 7 Mod 0, and Guide Mk 7 Mod 0. Figures 2-1, 2-2, and 2-3, and table 2-1 identify the major components and other parts of the launcher.

2-2. STAND MK 107 MOD 0

Table 2-1. (C) Major Units

Unit	Overall Dimensions (In.)[1] Height	Width	Depth	(cu ft)	(lbs)
Stand Mk 107 Mod 0	11⅜	16½	22½	2.4	110
Carriage Mk 7 Mod 0	8⅛	20	12	2.0	174

[1]Dimensions include mounting brackets.

2-3. (C) The stand (See figure 2-4) supports the launcher on bearings which permit rotation in train

Hoses and electrical cables enter the launcher through the stand by means of the cable and loop arrangement.

2-4. CARRIAGE MK 7 MOD 0

2-5. (C) The carriage (See figure 2-5) includes all trainable parts of the launcher with the exception of the guides and lower base ring.

2-6. (C) **Carriage Structure.** The carriage structural members (See figure 2-6) support the function components.

2-7. (C) TORQUE TUBE SUPPORTS. The torque tube supports (See figure 2-6) are equipped with bearings in which the torque tube rotates.

2-8. (C) **Shear Panels.** Two shear panels reinforce the torque tube supports. One panel is attached to the upper base ring and extends upward on the back of the supports. The other panel is attached to the guide support beam and to the top of the torque tube support

Section 2-2. Power Control Panel Mk 198 Mod 0

2-9. POWER CONTROL PANEL CABINET

2-10. (C) The power control panel cabinet consists of two cast aluminum housings, equipped with doors and seals, to form protective enclosures for the wiring and electrical components. The assembled cabinet is supported top and bottom by shock mounts.

2-11. (C) **Indicator Lights.** Twelve indicator light assemblies are mounted on the power control panel. Each light assembly consists of a double-socket base containing two lamps covered by a colored plexiglass dome.

Figure 3 (Sheet 1). Sample Page of Typeset Copy. (Par. 3.4.3)

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SECTION NO. AND TITLE USED AS SUBORDINATE BREAKDOWN WITHIN A CHAPTER PAR. 3.2.2.4.3.1

CHAPTER 2

DESCRIPTION

Section 2-1. Launcher Mk 112 Mod 0

2-1. (C) Launcher Mk 112 Mod 0 consists of Stand Mk 107 Mod 0, Carriage Mk 7 Mod 0, and Guide Mk 7 Mod 0. Figures 2-1, 2-2, and 2-3, and table 2-1 identify the major components and other parts of the launcher.

2-2. STAND MK 107 MOD 0

TABLE NO. AND TITLE PAR. 3.2.2.4.7.9

Table 2-1. (C) Major Units

Unit	Overall Dimensions (In.)[1] Height	Width	Depth	(cu ft)	(lbs)
Stand Mk 107 Mod 0	11⅜	16½	22½	2.4	110
Carriage Mk 7 Mod 0	8⅛	20	12	2.0	174

FOOTNOTE PAR. 3.1.15

[1]Dimensions include mounting brackets.

2-3. (C) The stand (See figure 2-4) supports the launcher on bearings which permit rotation in train. Hoses and electrical cables enter the launcher through the stand by means of the cable and loop arrangement.

2-4. CARRIAGE MK 7 MOD 0

2-5. (C) *The carriage (See figure 2-5)* includes all trainable parts of the launcher with the exception of the guides and lower base ring.

2-6. (C) Carriage Structure. The carriage structural members (See figure 2-6) support the function components.

FOURTH ORDER HEAD FIG. 1

2-7. (C) TORQUE TUBE SUPPORTS. The torque tube supports (See figure 2-6) are equipped with bearings in which the torque tube rotates.

2-8. (C) Shear Panels. Two shear panels reinforce the torque tube supports. One panel is attached to the upper base ring and extends upward on the back of the supports. The other panel is attached to the guide support beam and to the top of the torque tube support.

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Section 2-2. Power Control Panel Mk 198 Mod 0

2-9. POWER CONTROL PANEL CABINET

2-10. (C) The power control panel cabinet consists of two cast aluminum housings, equipped with doors and seals, to form protective enclosures for the wiring and electrical components. The assembled cabinet is supported top and bottom by shock mounts.

2-11. (C) Indicator Lights. Twelve indicator light assemblies are mounted on the power control panel. Each light assembly consists of a double-socket base containing two lamps covered by a colored plexiglass dome.

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Figure 3 (Sheet 2). Sample Page of Typeset Copy. (Par. 3.4.3)

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Figure 1-45. Fuel Transfer Level Control Valves Flow Diagram.

regulator is identical to that of the sump cell air pressure regulator. (See figure 1-66.)

1-106. **Bomb Bay Fuel Vent Shutoff Valve.** The valve is a normally open, pneumatically operated, is permitted to open the valve poppet. This action closes the bomb bay tank off from the main fuel vent line and permits the bomb bay tanks to be pressurized. A small poppet is incorporated to regulate the air into the piston chamber, preventing damage from high pressures vent line to allow the valve to open.

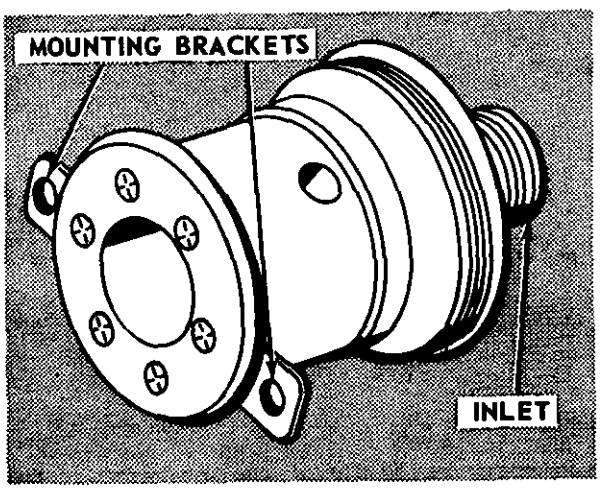


Figure 1-46. Saddle Tank Fuel Transfer Level Control Valve, BUNO 145157 through 146695.

1-107. **Bomb Bay Fuel Vent Relief Valve.** The vent relief valve is installed to relieve any excessive pressures in the bomb bay fuel tanks. It is a spring-loaded, popper-type relief valve, set to relieve pressures at a differential of 27 (±1) psi. (See figure 1-68.)

1-108. **Bomb Bay Tank Empty Float Switch.** The switch is located in the forward tank of either bomb bay configuration. When the tank contains fuel, the float is up, energizing the small internal relay in the switch and preventing the fuel transfer relay from being energized. The bomb bay fuel transfer air valve cannot close to prevent tank pressurization when the fuel transfer relay is deenergized, but the valve closes to prevent pressurization when the relay is energized. When the tank is dry, the float is down, establishing circuits to energize the fuel transfer air valve. This shuts off bomb bay tank pressurization.

Figure 3 (Sheet 3). Sample Page of Typeset Copy. (Par. 3.4.3)

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PAR. 3.1.5.2.1b, 3.4.6.3

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TECHNICAL MANUAL NO. PAR. 3.2.2.-4.1, 3.4.6.1 3.4.6.3

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FIGURE NO. AND TITLE PAR. 3.2.2.4.7.1 3.2.2.4.7.2 3.2.2.4.7.3 → *Figure 1-45. Fuel Transfer Level Control Valves Flow Diagram.*

regulator is identical to that of the sump cell air pressure regulator. (See figure 1-66.)

1-106. **Bomb Bay Fuel Vent Shutoff Valve.** The valve is a normally open, pneumatically operated, is permitted to operate the valve closed. This action closes the bomb bay tanks off from the main fuel vent line and permits the bomb bay tanks to be pressurized. A small popper is incorporated to regulate the air into the piston chamber, preventing damage from high pressures vent line to allow the valve to open.

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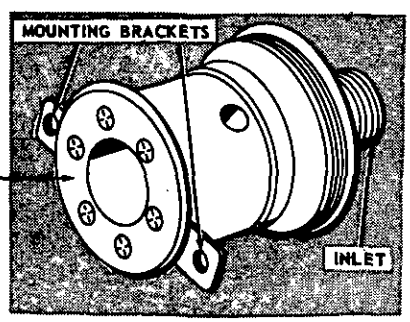


Figure 1-46. Saddle Tank Fuel Transfer Level Control Valve, BUNO 145157 through 146695.

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Figure 3 (Sheet 4). Sample Page of Typeset Copy. (Par. 3.4.3)

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CHAPTER 6

MAINTENANCE AND OVERHAUL

6–1. MAINTENANCE

6–2. Frequently inspect components of the system for visible signs of deterioration and defective parts. Replace or repair broken or damaged protective seals, and electrical and arming cables. Adjustments that do not require major disassembly are described in paragraph 5–6.

6–3. **Lubrication.** Lubricate the launching rack periodically in accordance with figure 6–1.

6–4. LAUNCHING CYLINDER. Disassemble the launching cylinder annually.

6–5. SHAFT BEARINGS. Approximately every six months, repack the two shaft bearings with grease. (See figure 6–2.) Proceed as follows:

3. Remove nut from piston extension on forward side of power frame casing.

4. Force rail to extend 4 or 5 inches. Use small bar.

5. Place brass striking piece over threaded portion of tapered extension; rap piece sharply to unseat taper.

NOTE: If the taper is seized, and will not readily unseat, use a hydraulic jack.

6. Place supporting wood blocking under forward and aft ends of cylinder.

7 Remove four capscrews from bracket

6–9. Launching Rail Cylinder. Figure 6–3 illustrates completely disassembled cylinder. Figure 2–4 illustrates cylinder installed. Removed cylinder as follows:

1. Retract rail.

WARNING

Secure air supply to system and bleed pressure by extending and retracting snubbers to prevent uncontrolled movement from crushing operator. Attach appropriate WARNING tag.

2. Disconnect air lines to cylinder. Plug lines to prevent entrance of dirt or foreign matter.

3. Remove bracket and ... rods.

CAUTION

Note exact order in which parts are removed. Cushioning feature described in chapter 2 depends on correct assembly.

6–11. **Rail Control Valves.** Repair and overhaul instructions for these valves are identical with the elevation stow latch and guide drive pin control valves instructions.

6–12. **Temperature Sensing Device.** Remove the

Figure 3 (Sheet 5). Sample Page of Typeset Copy. (Par. 3.4.3)

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CHAPTER NO. AND TITLE PAR. 3.2.1.13; 3.2.2.4.2 3.2.2.5.6, 3.4.5.4

CHAPTER 6

MAINTENANCE AND OVERHAUL

FIRST ORDER HEAD FIG. 1

6–1. MAINTENANCE

6–2. Frequently inspect components of the system for visible signs of deterioration and defective parts. Replace or repair broken or damaged protective seals, and electrical and arming cables. Adjustments that do not require major disassembly are described in paragraph 5–6.

6–3. Lubrication. Lubricate the launching rack periodically in accordance with figure 6–1.

6–4. LAUNCHING CYLINDER. Disassemble the launching cylinder annually.

6–5. SHAFT BEARINGS. Approximately every six months, repack the two shaft bearings with grease. (See figure 6–2.) Proceed as follows:

3. Remove nut from piston extension on forward side of power frame casing.

4. Force rail to extend 4 or 5 inches. Use small bar.

5. Place brass striking piece over threaded portion of tapered extension; rap piece sharply to unseat taper.

NOTE: If the taper is seized, and will not readily unseat, use a hydraulic jack.

STEPS OF PROCEDURE PAR. 3.2.2.4.5.6, 3.4.5.2

6. Place supporting wood blocking under forward and aft ends of cylinder.

7. Remove four capscrews from bracket

6–9. Launching Rail Cylinder. Figure 6–5 illustrates completely disassembled cylinder. Figure 2–4 illustrates cylinder installed. Removed cylinder as follows:

1. Retract rail.

WARNING PAR. 3.1.14 3.4.5.2

WARNING

Secure air supply to system and bleed pressure by extending and retracting snubbers to prevent uncontrolled movement from crushing operator. Attach appropriate WARNING tag.

2. Disconnect air lines to cylinder. Plug lines to prevent entrance of dirt or foreign matter.

3. Remove bracket and tie rods.

CAUTION PAR. 3.1.14 3.4.5.2

CAUTION

Note exact order in which parts are removed. Cushioning feature described in chapter 2 depends on correct assembly.

6–11. Rail Control Valves. Repair and overhaul instructions for these valves are identical with the elevation stow latch and guide drive pin control valves instructions.

6–12. Temperature Sensing Device. Remove the

Figure 3 (Sheet 6). Sample Page of Typeset Copy. (Par. 3.4.3)

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SECTION VI
ENGINE SECTION

6-1. ENGINE SECTION

6-2. The engine section (See figure 6-1) includes the fixed nose section, nose flaps, an antidrag ring, cowl bow, cowl flaps, upper and lower accessory cowling panels, carburetor air scoop, oil cooler ducts, oil cooler supporting structure, and the engine mount.

6-3. Sealing. Certain areas of the engine section are weather-and-fume sealed. Applied sealants (organic compounds) and mechanical seals (rubber, metal, or synthetic) must be replaced when destroyed or removed for structural repair. All structural repair members (splice-plates, patches, fillers) must be sealed when installed in sealed areas. Refer to section X.

6-4. Corrosion Inspection. When making structural repairs to any component or area of the engine section, an inspection must also be made for corrosion in the damaged area and the adjacent area. Refer to section IX.

NOTE: This corrosion inspection augments, but does not replace, the periodic inspection made during [illegible] maintenance schedules.

forward channel (0.0[illegible]) and the aft web (0.040) are made from Al 2024-T Alclad. Miscellaneous structure includes gussets (0.050 Al 2024-T Alclad) and brackets (0.063 Al 2024-T Alclad). The structure of the fixed nose fairing consists of three doublers, all of 0.040 Al 6061-T alloy. The fixed nose fairing is attached to the engine with four channels, all of 0.063 Al 6061-T alloy.

6-8. NEGLIGIBLE DAMAGE. Smooth dents and abrasions free of cracks and scratches which do not penetrate the clad surface may be disregarded. Smoothed nicks or stop-drilled cracks (use No. 40—0.098 drill) in the edges of the bulkhead flanges, gusset flanges, web flanges, or lightening hole flanges which do not exceed ⅛ inch, and which are at least four rivet diameters (4D) from existing holes and at least two inches from any adjacent damage, may be classified as negligible damage. The damage must be trimmed, smoothed, stop-drilled, or burnished before a negligible damage classification is made. The only damage to the hook-shaped extrusion that may be considered negligible is damage consisting of shallow defects not exceeding 1/16 inch in depth. Paint all negligible damage areas with at least two coats of zinc chromate primer as a corrosion inhibitor.

6-9. DAMAGE REPAIRABLE BY PATCHING. Damage to a lightening hole flange may be repaired with flanged doublers attached to the web or bulkhead with AD4 (⅛ inch) rivets spaced ¾ inch apart. If the crack is near a gusset, the doubler must be installed under the gusset and should catch all existing rivets. (See figure B-15.) Elongated rivet holes must be repaired by installing a doubler. Use existing rivet holes to locate rivet holes in the doubler and use the same specification-type rivet as

[illegible] section must incorporate contour checks during and after repairs.

6-10. DAMAGE NECESSITATING REPLACEMENT. When a structural component of the fixed nose section has been damaged beyond repair, or when replacement is more readily accomplished or more economical than repair, the structural component should be replaced. The replacement part must be of the same gage and material as the original, and should be attached with the same rivet pattern.

Figure 3 (Sheet 7). Sample Page of Typeset Copy. (Par. 3.4.3)

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SECTION VI

ENGINE SECTION

SECTION NO. AND TITLE PAR. 3.2.1.13, 3.2.2.4.3.2 3.2.2.5.6, 3.4.5.4

FIRST ORDER HEAD — FIG. 1

6-1. ENGINE SECTION

6-2. The engine section (see figure 6-1) includes the fixed nose section, nose flaps, an antidrag ring, cowl bow, cowl flaps, upper and lower accessory cowling panels, carburetor air scoop, oil cooler ducts, oil cooler supporting structure, and the engine mount.

SECOND ORDER HEAD — FIG. 1

6-3. Sealing. Certain areas of the engine section are weather-and-fume sealed. Applied sealants (organic compounds) and mechanical seals (rubber, metal, or synthetic) must be replaced when destroyed or removed for structural repair. All structural repair members (splice-plates, patches, fillers) must be sealed when installed in sealed areas. Refer to section X.

6-4. Corrosion Inspection. When making structural repairs to any component or area of the engine section, an inspection must also be made for corrosion in the damaged area and the adjacent area. Refer to section IX.

NOTE 3.1.14 3.4.5.2

NOTE: This corrosion inspection augments, but does not replace, the periodic inspection [illegible] regular [illegible] schedule.

[illegible] and channels [illegible] and the air [illegible] are made from Al 2024-T Alclad. Miscellaneous structure includes gussets (0.050 Al 2024-T Alclad) and brackets (0.063 Al 2024-T Alclad). The structure of the fixed nose fairing consists of three doublers, all of 0.040 Al 6061-T alloy. The fixed nose fairing is attached to the engine with four channels, all of 0.063 Al 6061-T alloy.

THIRD ORDER HEAD — FIG. 1

6-8. NEGLIGIBLE DAMAGE. Smooth dents and abrasions free of cracks and scratches which do not penetrate the clad surface may be disregarded. Smoothed nicks or stop-drilled cracks (use No. 40—0.098 drill) in the edges of the bulkhead flanges, gusset flanges, web flanges, or lightening hole flanges which do not exceed ⅛ inch, and which are at least four rivet diameters (4D) from existing holes and at least two inches from any adjacent damage, may be classified as negligible damage. The damage must be trimmed, smoothed, stop-drilled, or burnished before a negligible damage classification is made. The only damage to the hook-shaped extrusion that may be considered negligible is damage consisting of shallow defects not exceeding 1/16 inch in depth. Paint all negligible damage areas with at least two coats of zinc chromate primer as a corrosion inhibitor.

6-9. DAMAGE REPAIRABLE BY PATCHING. Damage to a lightening hole flange may be repaired with flanged doublers attached to the web or bulkhead with AD4 (⅛ inch) rivets spaced ¾ inch apart. If the crack is near a gusset, the doubler must be installed under the gusset and should catch all existing rivets. (See figure B-15.) Elongated rivet holes must be repaired by installing a doubler. Use existing rivet holes to locate rivet holes in the doubler [illegible] the same specification rivet as [illegible] section [illegible] separate co[illegible]ks during and after repairs.

6-10. DAMAGE NECESSITATING REPLACEMENT. When a structural component of the fixed nose section has been damaged beyond repair, or when replacement is more readily accomplished or more economical than repair, the structural component should be replaced. The replacement part must be of the same gage and material as the original, and should be attached with the same rivet pattern.

6-1

MARGINAL COPY FOR RIGHT-HAND PAGE PAR. 3.2.2.4.4 3.2.2.4.4.3 3.4.6.3

Figure 3 (Sheet 8). Sample Page of Typeset Copy. (Par. 3.4.3)

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FOREWORD

Ordnance Pamphlet 0000 (First Revision) (Volume 1) describes Attack Director Mk 00 Mod 0, and provides instructions for operation.

This publication consists of four volumes.

Volume 1 - Description and Operation
- Chapter 1 - Introduction
- Chapter 2 - Physical Description
- Chapter 3 - Functional Description
- Chapter 4 - Operation

Volume 2 - Maintenance
- Chapter 5 - Routine Maintenance
- Chapter 6 - Preventive Maintenance
- Chapter 7 - Troubleshooting

Volume 3 - Functional Schematics and Wiring Diagrams
- Chapter 8 - Functional Schematics and Wiring Diagrams

Volume 4 - Replaceable Parts List
- Chapter 9 - Replaceable Parts List

FOREWORD

Figure 4. Sample FOREWORD. (Par. 3.2.1.7)

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SAFETY SUMMARY

The following are general safety precautions that are not related to any specific procedures and therefore do not appear elsewhere in this publication. These precautions are recommended safety precautions that personnel must understand and apply during many of the phases of operation and maintenance.

KEEP AWAY FROM LIVE CIRCUITS

Operating personnel must at all times observe all safety regulations. Do not change tubes or make adjustments inside equipment with high voltage supply on. Under certain conditions dangerous potentials may exist in circuits with power control in the off position because of charges retained by capacitors. To avoid casualties always remove power and discharge and ground circuits prior to touching them.

DO NOT SERVICE OR ADJUST ALONE

Under no circumstances should any person reach within or enter the enclosure for the purpose of servicing or adjusting the equipment without the immediate presence or assistance of another person capable of rendering aid.

RESUSCITATION

Personnel working with or near high voltages should be familiar with the methods of artificial respiration. Such information should be obtained from the Bureau of Medicine and Surgery.

The following WARNINGS are repeated from the text for the protection of personnel.

Figure 5 (Sheet 1). Sample SAFETY SUMMARY. (Par. 3.2.1.12)

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WARNINGS

The slip ring assembly contains exposed electrical components operating at dangerous voltages. All power to ring assembly must be disconnected before any doors or access covers are removed. (Page 6-3)

Be sure that all personnel are clear of the telemetering antennas before moving the antenna. (Page 8-10)

During loading operations, be sure that the stop bolt of the stop mechanism is in the LOAD position. If the stop bolt is not in this position, the torpedo could strike the muzzle door. (Page 8-42)

The following major CAUTIONS are repeated from the text because if not strictly observed the effectiveness of the equipment or ship may be destroyed.

CAUTIONS

When on-loading missiles, take special care to set the switches correctly. An improperly stowed missile may result in a time delay that endangers the ship. (Page 6-24)

Depressing the SYSTEM RESET pushbutton while a missile is in flight and before the KILLED or SURVIVED indicator lamp lights will cause the missile to be destroyed before it engages the target. (Page 6-30)

Figure 5 (Sheet 2). Sample SAFETY SUMMARY. (Par. 3.2.1.12)

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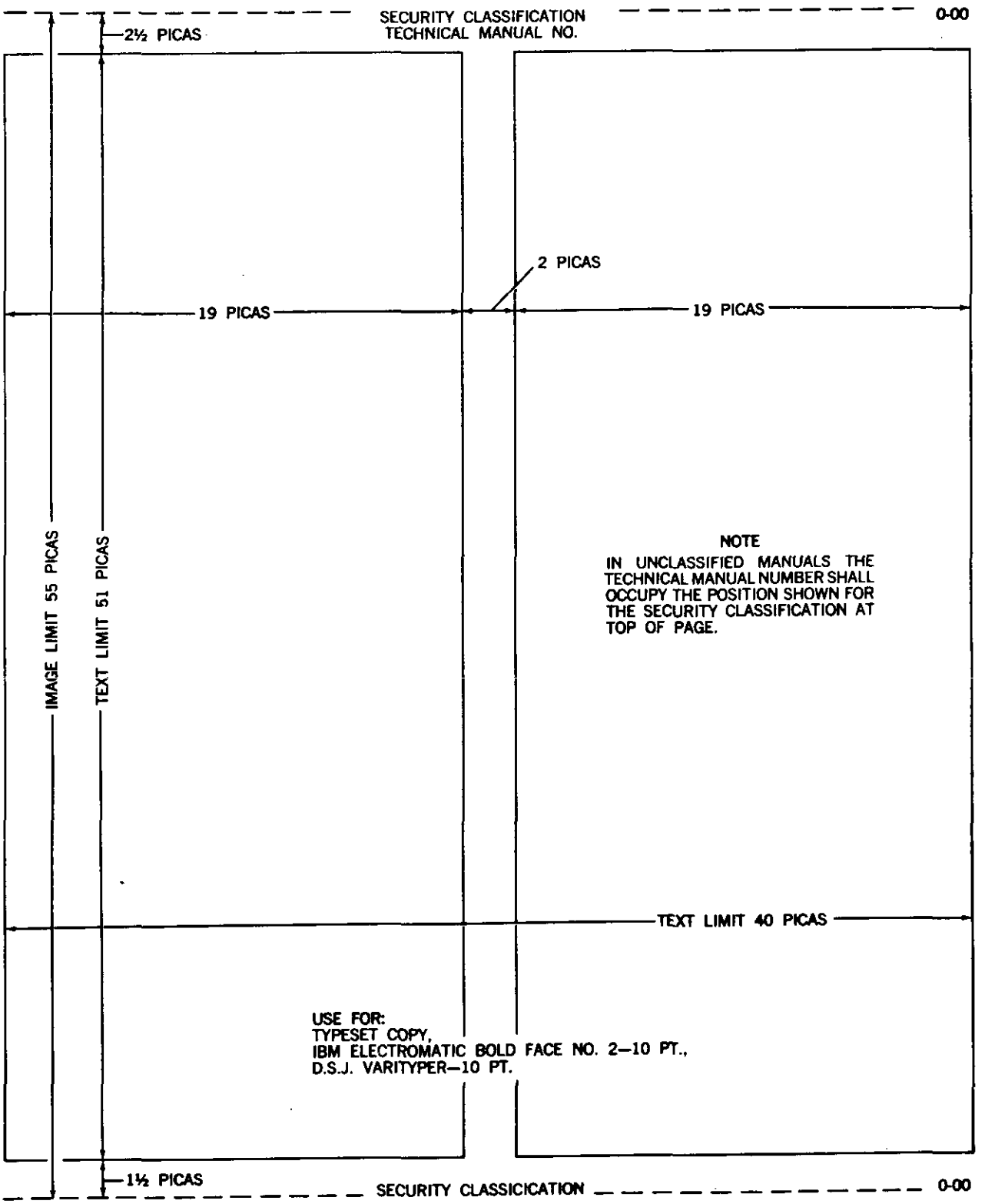


Figure 6. Dimensions for Reproduction Size Image Area. (Par. 3.2.2.1.1)

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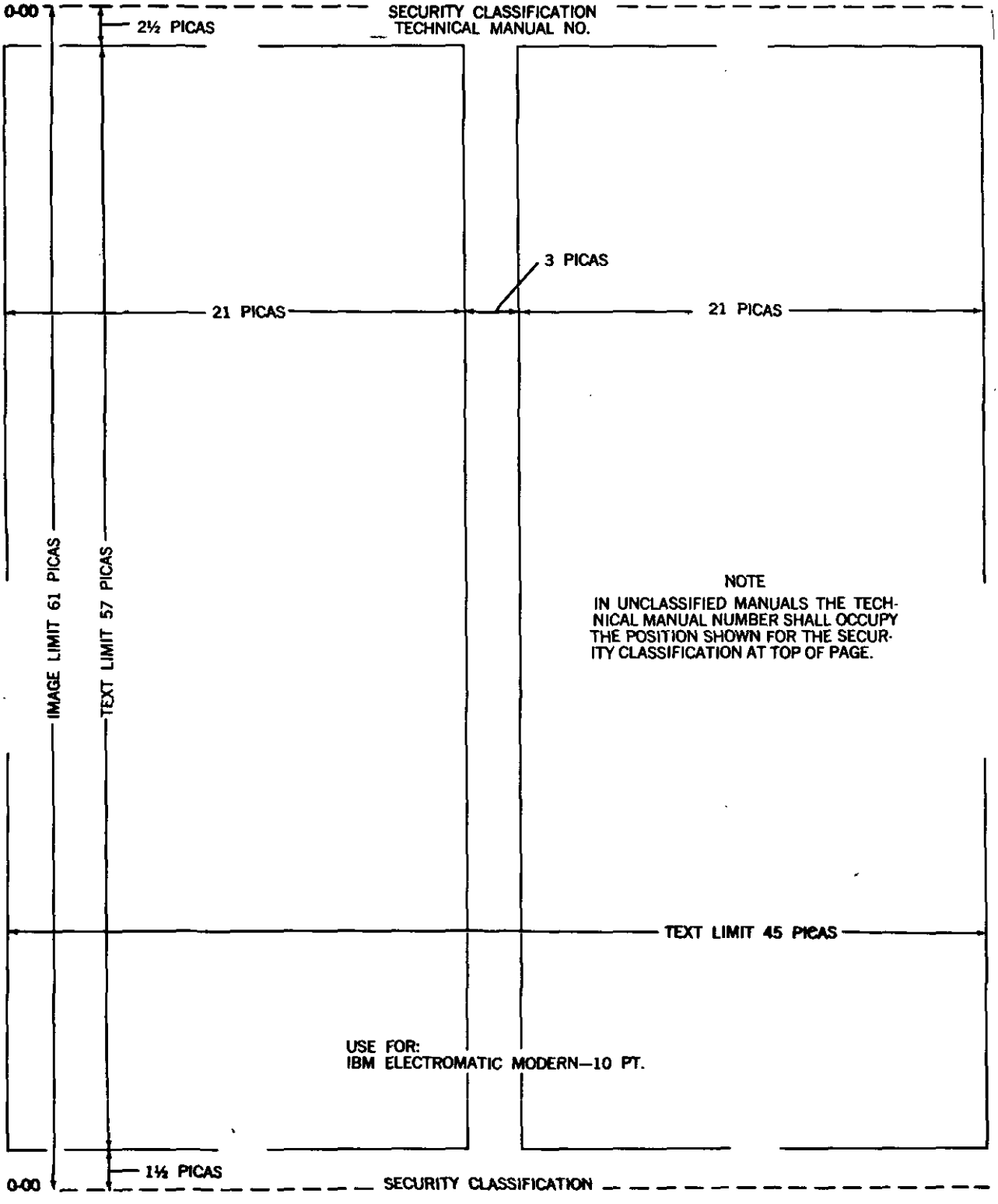


Figure 7. Dimensions for 10-Percent Oversize Image Area. (Par. 3.2.2.1.2)

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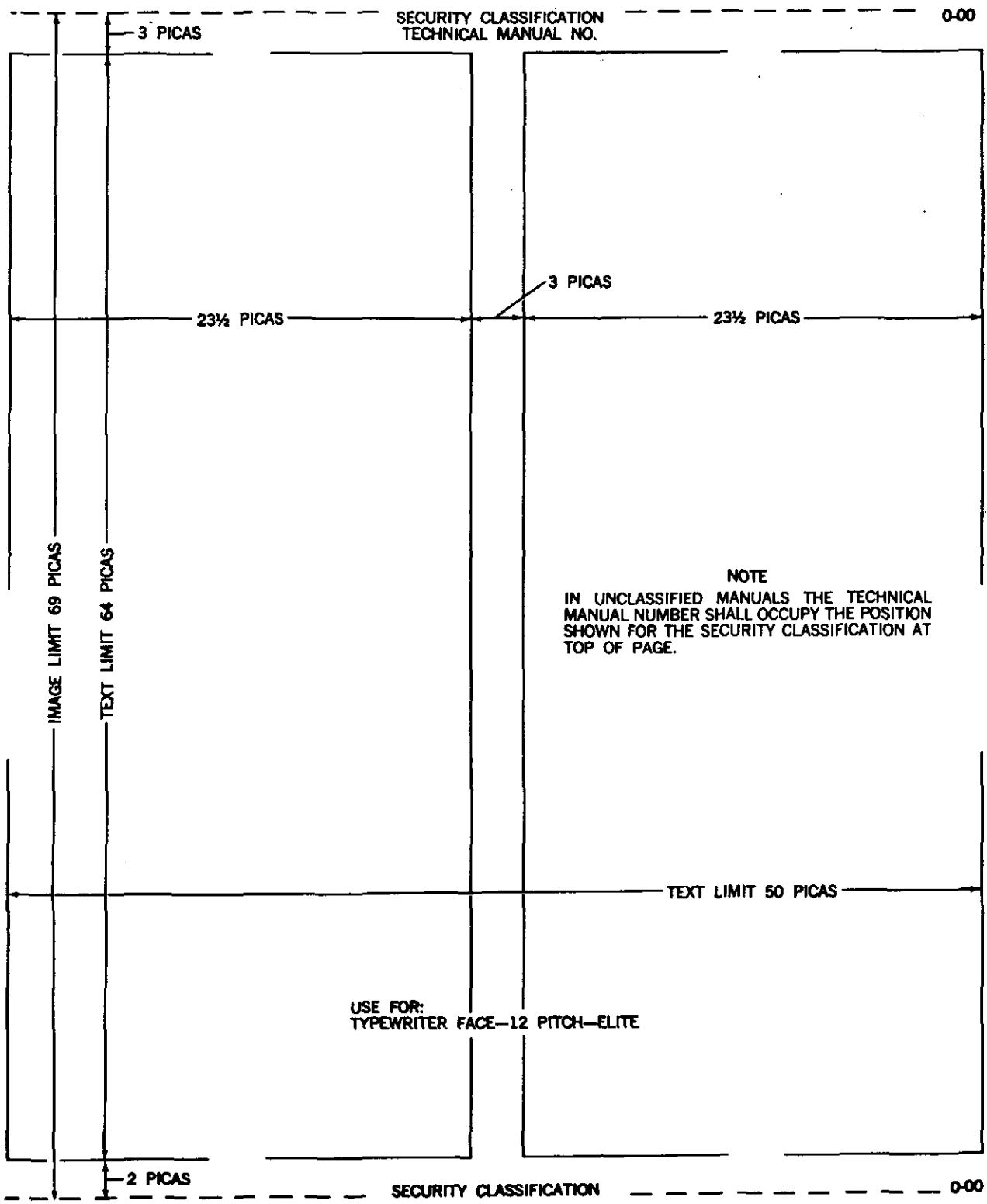


Figure 8. Dimensions for 20-Percent Oversize Image Area. (Par. 3.2.2.1.2)

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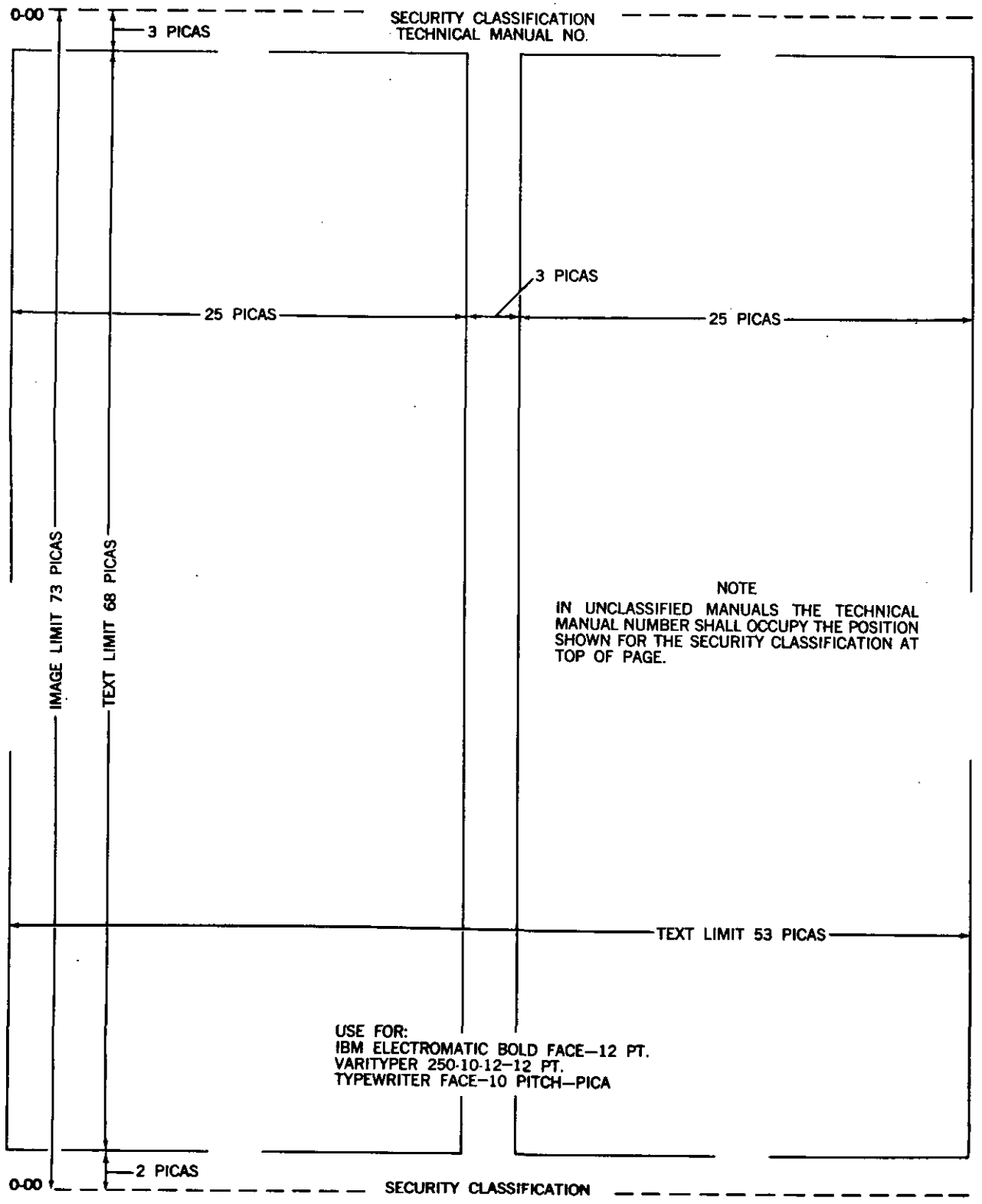


Figure 9. Dimensions for 25-Percent Oversize Image Area. (Par. 3.2.2.1.2)

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CHAPTER 2

DESCRIPTION

Section 2-1. Launcher Mk 112 Mod 0

2-1. (C) Launcher Mk 112 Mod 0 consists of Stand Mk 107 Mod 0, Carriage Mk 7 Mod 0, and Guide Mk 7 Mod 0. Figures 2-1, 2-2, and 2-3 and table 2-1 identify the major components and other parts of the launcher.

Table 2-1. (C) Major Units

Unit	Overall Dimensions (in.)[1]			(cu ft)	(lbs)
	Height	Width	Depth		
Stand Mk 107 Mod 9	11⅜	16½	22½	2.4	100
Carriage Mk 7 Mod 0	8⅛	20	12	2.0	174

[1] Dimensions include mounting brackets.

2.2. STAND MK 7 MOD 0

2-4. The stand (see figure 2-4) supports

2-5. (C) The carriage (see figure 2-5) includes all trainable parts of the launcher with the exception of the guides and lower base ring.

2-6. (C) CARRIAGE STRUCTURE. The carriage structural members (see figure 2-6) support the function components.

2-7. (C) TORQUE TUBE SUPPORTS. The torque tube supports (see figure 2-6) are equipped with bearings in which the torque tube rotates.

2-8. (C) SHEAR PANELS. Two shear panels reinforce the torque tube supports. One panel is attached to the upper base ring and extends upward on the back of the supports. The other panel is attached to the guide support beam and to the top of the torque tube support.

Section 2-2. Power Control Panel Mk 198 Mod 0

2.9. POWER CONTROL PANEL CABINET

2-10. (C) The power control panel cabinet consists of two cast aluminum housings, equipped with doors and seals, to form protective enclosures for the wiring and electrical components. The assembled cabinet is supported top and bottom by shock mounts.

2-11. (C) INDICATOR LIGHTS. Twelve indicator

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Figure 10 (Sheet 1). Sample Page of DSJ Varitype Copy. (Par. 3.2.2.1.2.1)

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Figure 1-45. Fuel Transfer Level Control Valves Flow Diagram.

regulator is identical to that of the sump cell air pressure regulator. (See figure 1-66.)

1-106. BOMB BAY FUEL VENT SHUTOFF VALVE. The valve is a normally open, pneumatically operated, gate-type shutoff valve. (See figure 1-67.) When the bomb bay fuel transfer air valve is open, compressed air is permitted to operate the valve closed. This action closes the bomb bay tanks off from the main fuel vent line and

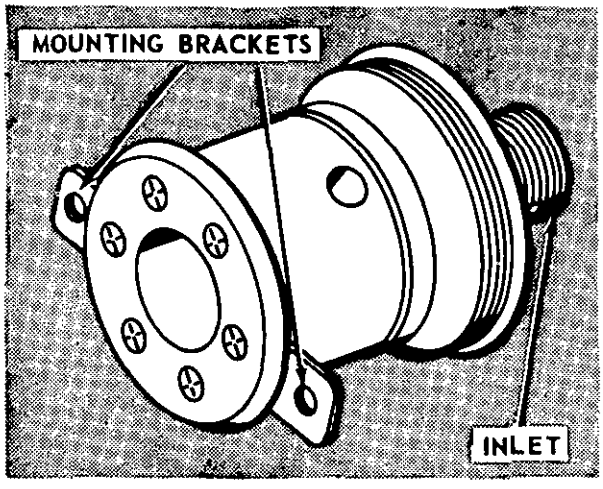


Figure 1-46. Saddle Tank Fuel Transfer Level Control Valve, BUNO 145157 through 146696.

shutoff valve is closed, air from the piston chamber is vented to the main vent line to allow the valve to open.

1-107. BOMB BAY FUEL VENT RELIEF VALVE. The vent relief valve is installed to relieve any excessive pressures in the bomb bay fuel tanks. It is a spring-loaded, poppet-type relief valve, set to relieve pressures at a differential of 27 (±1) psi. (See figure 1-68.)

1-108. BOMB BAY TANK EMPTY FLOAT SWITCH. The switch is located in the forward tank of either bomb bay configuration. When the tank contains fuel, the float is up, energizing the small internal relay in the switch and preventing the fuel transfer relay from being energized. The bomb bay fuel transfer air valve cannot close to prevent tank pressurization when the fuel transfer relay is de-

Figure 10 (Sheet 2). Sample Page of DSJ Varitype Copy. (Par. 3.2.2.1.2.1)

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CHAPTER 6

MAINTENANCE AND OVERHAUL

6-1. MAINTENANCE

6-2. Frequently inspect components of the system for visible signs of deterioration and defective parts. Replace or repair broken- or damaged protective seals, and electrical and arming cables. Adjustments that do not require major disassembly are described in paragraph 5-6.

6-3. LUBRICATION. Lubricate the launching rack periodically in accordance with figure 6-1.

6-4. LAUNCHING CYLINDER. Disassemble the launching cylinder annually.

6-5. SHAFT BEARINGS. Approximately every six months, repack the two shaft bearings with grease. (See fi... 6-2.) Proceed as follows:

2. Disconnect air lines to cylinder. Plug lines to prevent entrance of dirt or foreign matter.

3. Remove nut from piston extension on forward side of power frame casing.

4. Force rail to extend 4 or 5 inches. Use small bar.

5. Place brass-striking piece over threaded portion of tapered extension; rap piece sharply to unseat taper.

NOTE: If the taper is seized, and will not readily unseat, use a hydraulic jack.

6. Place supporting wood blocking under forward and aft ends of cylinder.

7. Remove four cap screws from bracket.

6-6. LAUNCHING RAIL CYLINDER. Figure 6-3 illustrates completely disassembled cylinder. Figure 2-4 illustrates cylinder installed. Remove cylinder as follows:

1. Retract rail.

WARNING

Secure air supply to system and bleed pressure by extending and retracting snubbers to prevent uncontrolled movement from crushing operator. Attach appropriate WARNING tag.

8. Remove bracket and tie rods.

CAUTION

Note exact order in which parts are removed. Cushioning feature described in chapter 2 depends on correct assembly.

6-11. Rail Control Valves. Repair and overhaul instructions for these valves are identical with the elevation stow latch and guide drive pin control valves instructions.

6-12. Temperature Sensing Device. Remove the

Figure 10 (Sheet 3). Sample Page of DSJ Varitype Copy. (Par. 3.2.2.1.2.1)

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SECTION VI

ENGINE SECTION

6-1. ENGINE SECTION

6-2. The engine section (see figure 6-1) includes the fixed nose section, nose flaps, an antidrag ring, cowl bow, cowl flaps, upper and lower accessory cowling panels, carburetor air scoop, oil cooler ducts, oil cooler supporting structure, and the engine mount.

6-3. SEALING. Certain areas of the engine section are weather-and-fume sealed. Applied sealants (organic compounds) and mechanical seals (rubber, metal, or synthetic) must be replaced when destroyed or removed for structural repair. All structural repair members (splice-plates, patches, fillers) must be sealed when installed in sealed areas. Refer to section X.

6-4. CORROSION INSPECTION. When making structural repairs to any component or area of the engine section, an inspection must also be made for corrosion in the damaged area and the adjacent area. Refer to section IX.

NOTE: This corrosion inspection augments, but does not replace, the periodic inspection made during regular maintenance schedules.

6-7. [illegible] (See figure [illegible] structure of the fixed nose ring consists of bulkheads (or ribs), a forward channel, an aft web, gussets, and brackets. The bulkheads (or ribs) extend from the forward tip of the nose ring to the aft web, and are made of 0.040 Al 2024-T Alclad. The forward channel (0.050) and the aft web (0.040) are made from Al 2024-T Alclad. Miscellaneous structure includes gussets (0.050 Al 2024-T Alclad) and brackets (0.063 Al 2024-T Alclad). The structure of the fixed nose fairing consists of three doublers, all of 0.040 Al 6060-T alloy. The fixed nose fairing is attached to the engine with four channels, all of 0.063 Al 6061-T alloy.

6-8. NEGLIGIBLE DAMAGE. Smooth dents and abrasions free of cracks and scratches which do not penetrate the clad surface may be disregarded. Smoothed nicks or stop-drilled cracks (use No. 40–.098 drill) in the edges of the bulkhead flanges, gusset flanges, web flanges, or lightening hole flanges which do not exceed 1/8 inch, and which are at least four rivet diameters (4D) from existing holes and at least two inches, from any adjacent damage, may be classified as negligible damage. The damage must be trimmed, smoothed, stop-drilled, or burnished before a negligible damage classification is made. The only damage to the hook-shaped extrusion that may be considered [illegible]ible is dam[illegible]sing of shallo[illegible]

[illegible]ged doublers [illegible] web or bulkhead with AD4 (1/8 inch) rivets spaced 3/4 inch apart. If the crack is near a gusset, the doubler must be installed under the gusset and

6-1

Figure 10 (Sheet 4). Sample Page of DSJ Varitype Copy. (Par. 3.2.2.1.2.1)

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CHAPTER 2

DESCRIPTION

Section 2-1. Launcher Mk 112 Mod 0

2-1. (C) Launcher Mk 112 Mod 0 consists of Stand Mk 107 Mod 0, Carriage Mk 7 Mod 0, and Guide Mk 7 Mod 0. Figures 2-1, 2-2, and 2-3, and table 2-1 identify the major components and other parts of the launcher.

2-2. STAND MK 107 MOD 0

TABLE 2-1. (C) MAJOR UNITS

Unit	Overall Dimensions (in.)*			(Cu Ft)	(Lbs)
	Height	Width	Depth		
Stand Mk 107 Mod 0	11 3/8	16 1/2	22 1/2	2.4	110
Carriage Mk 7 Mod 0	8 1/8	20	12	2.0	174

...ensions ...de ...ting ...kets

2-3. (C) The stand (see figure 2-4) supports the launcher on bearings which permit rotation in train. Hoses and electrical cables enter the launcher through the stand by means of the cable and loop arrangement.

2-4. CARRIAGE MK 7 MOD 0

2-5. The carriage (see figure 2-5) includes all trainable parts of the launcher with the exception of the guides and lower base ring.

2-6. (C) CARRIAGE STRUCTURE. The carriage structural members (see figure 2-6) support the function components.

2-7. (C) Torque Tube Supports. The torque tube supports (see figure 2-6) are equipped with bearings in which the torque tube rotates.

2-8. (C) Shear Panels. Two shear ...els reinfo... ...rque tub... ...orts

Section 2-2. Power Control Panel Mk 198 Mod 0

2-9. (C) POWER CONTROL PANEL CABINET

2-10. (C) The power control panel cabinet consists of two cast aluminum housings, equipped with doors and seals, to form protective enclosures for the wiring and electrical components. The assembled cabinet is supported top and bottom by shock mounts.

2-11. (C) INDICATOR LIGHTS. Twelve indicator light assemblies are mounted on the power control panel. Each light assembly consists of a double-socket base containing two lamps covered by a

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Figure 11 (Sheet 1). Sample Page of IBM Electromatic Copy. (Par. 3.2.2.1.2.1)

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Figure 1-45. Fuel Transfer Level Control Valves Flow Diagram.

regulator is identical to that of the sump cell air pressure regulator. (See figure

air valve is open, compressed air is permitted to operate the valve closed. This action closes the bomb bay tanks off from

surges. When the bomb bay fuel vent shutoff valve is closed, air from the piston chamber is vented to the main vent line to allow the valve to open.

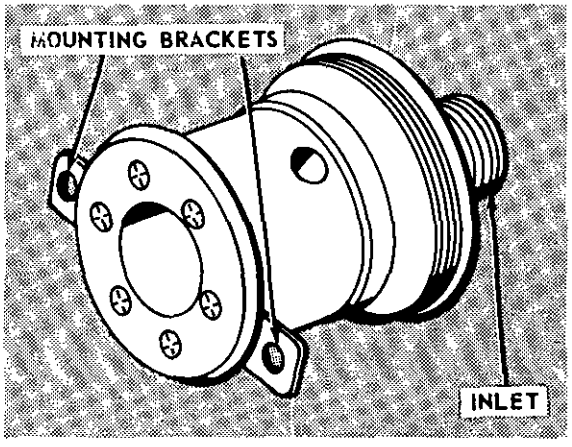


Figure 1-46. Saddle Tank Fuel Transfer Level Control Valve, BUNO 145157 through 146696.

1-107. BOMB BAY FUEL VENT RELIEF VALVE. The vent relief valve is installed to relieve any excessive pressures in the bomb bay fuel tanks. It is a spring-loaded, poppet-type relief valve, set to relieve pressures at a differential of 27 (±1) psi. (See figure 1-68.)

1-108. BOMB BAY TANK EMPTY FLOAT SWITCH. The switch is located in the forward tank of either bomb bay configuration. When the tank contains fuel, the float is up, energizing the small internal relay in the switch and preventing the fuel transfer relay from being energized. The bomb bay fuel transfer air

Figure 11 (Sheet 2). Sample Page of IBM Electromatic Copy. (Par. 3.2.2.1.2.1)

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CHAPTER 6

MAINTENANCE AND OVERHAUL

6-1. MAINTENANCE

6-2. Frequently inspect components of the system for visible signs of deterioration and defective parts. Replace or repair broken or damaged protective seals, and electrical and arming cables. Adjustments that do not require major disassembly are described in paragraph 5-6.

6-3. LUBRICATION. Lubricate the launching rack periodically in accordance with figure 6-1.

6-4. Launching Cylinder. Disassemble the launching cylinder annually.

6-5. Shaft Bearings. Approximately

3. Remove nut from piston extension on forward side of power frame casing.

4. Force rail to extend 4 or 5 inches. Use small bar.

5. Place brass striking piece over threaded portion of tapered extension; rap piece sharply to unseat taper.

NOTE: If the taper is seized, and will not readily unseat, use a hydraulic jack.

6. Place supporting wood blocking under forward and aft ends of cylinder.

7. Remove four capscrews from

Proceed as follows:

1. Retract rail.

WARNING

Secure air supply to system and bleed pressure by extending and retracting snubbers to prevent uncontrolled movement from crushing operator. Attach appropriate WARNING tag.

2. Disconnect air lines to cylinder. Plug lines to prevent entrance of dirt or foreign matter.

CAUTION

Note exact order in which parts are removed. Cushioning feature described in chapter 2 depends on correct assembly.

6-11. RAIL CONTROL VALVES. Repair and overhaul instructions for these valves are identical with the elevation stow latch and guide drive pin control valves instructions.

6-12. TEMPERATURE SENSING

Figure 11 (Sheet 3). Sample Page of IBM Electromatic Copy. (Par. 3.2.2.1.2.1)

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SECTION VI

ENGINE SECTION

6-1. ENGINE SECTION

6-2. The engine section (see figure 6-1) includes the fixed nose section, nose flaps, an antidrag ring, cowl bow, cowl flaps, upper and lower accessory cowling panels, carburetor air scoop, oil cooler ducts, oil cooler supporting structure, and the engine mount.

6-3. SEALING. Certain areas of the engine section are weather-and-fume sealed. Applied sealants (organic compounds) and mechanical seals (rubber, metal, or synthetic) must be replaced when destroyed or removed for structural repair. All structural repair members (splice-plates, patches, fillers) must be sealed when installed in sealed areas. Refer to section X.

6-4. CORROSION INSPECTION. When making structural repairs to any component or area of the engine section, an inspection must also be made for corrosion in the damaged area and the adjacent area. Refer to section IX.

NOTE: This corrosion inspection augments, but does not replace, the periodic inspection made dur-

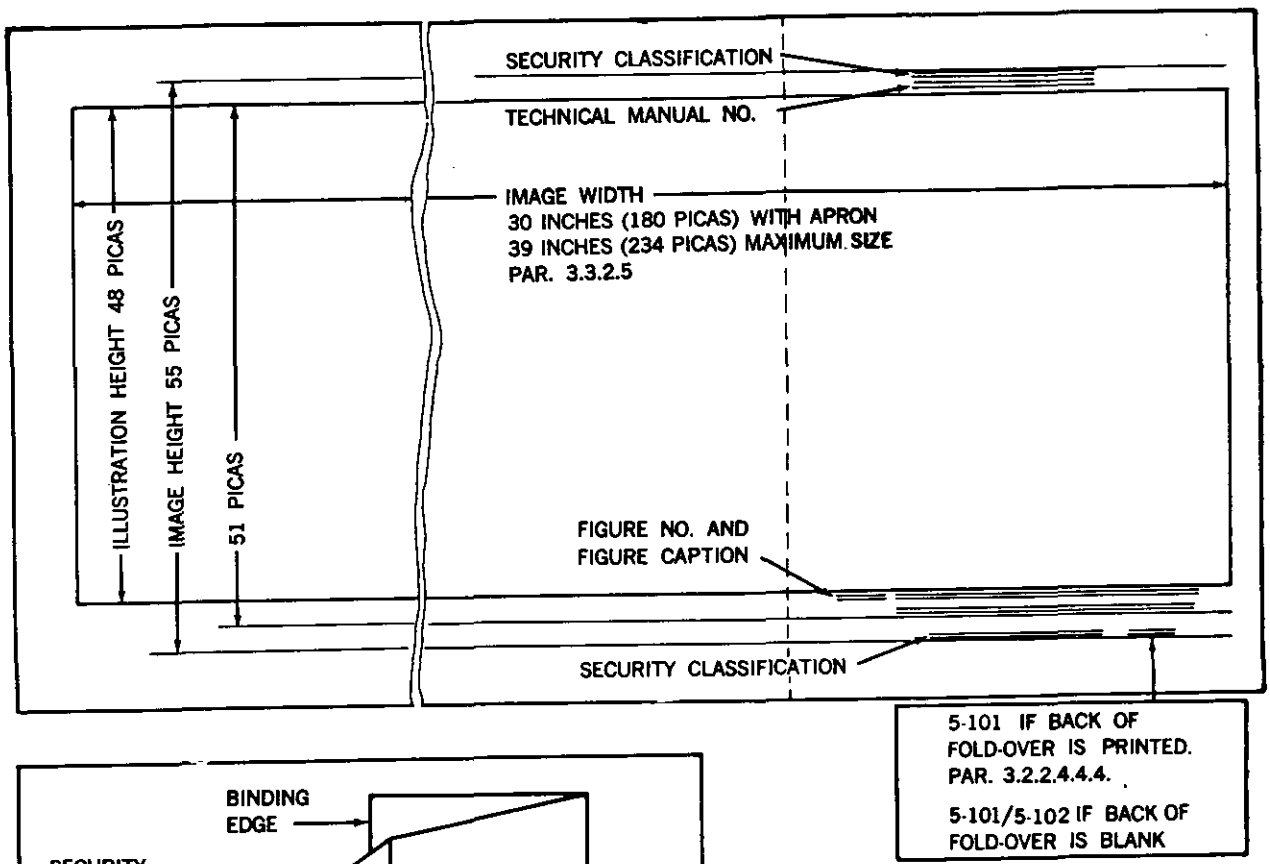
6-8. Negligible Damage. Smooth dents and abrasions free of cracks and scratches which do not penetrate the clad surface may be disregarded. Smoothed nicks or stop-drilled cracks (use No. 40—.098 drill) in the edges of the bulkhead flanges, gusset flanges, web flanges, or lightening hole flanges which do not exceed 1/8 inch and which are at least four rivet diameters (4D) from existing holes and at least two inches from any adjacent damage may be classified as negligible damage. The damage must be trimmed, smoothed, stop-drilled, or burnished before a negligible damage classification is made. The only damage to the hookshaped extrusion that may be considered negligible is damage consisting of shallow defects not exceeding 1/16 inch in depth. Paint all negligible damage areas with at least two coats of zinc chromate primer as a corrosion inhibitor.

6-9. Damage Repairable by Patching. Damage to a lightening hole flange may be repaired with flanged doublers attached to the web or bulkhead with AD4 (1/8 inch) rivets spaced 3/4 inch apart. If the crack is near a gusset, the doubler must be installed under the gusset and should catch all existing rivets, figure B-15. Elongated rivet holes must be repaired by installing a doubler. Use existing rivet holes to locate rivet holes in the doubler and use the same specification-type rivet as those used in the original structure. Repair to an extensively damaged flange may necessitate cutting away the damaged flange and in-

6-10. Damage Necessitating Replacement. When a structural component of the fixed nose section has been damaged beyond repair, or when replacement is more readily accomplished or more economical than repair, the structural component should be replaced. The replacement part must be of the same

Figure 11 (Sheet 4). Sample Page of IBM Electromatic Copy. (Par. 3.2.2.1.2.1)

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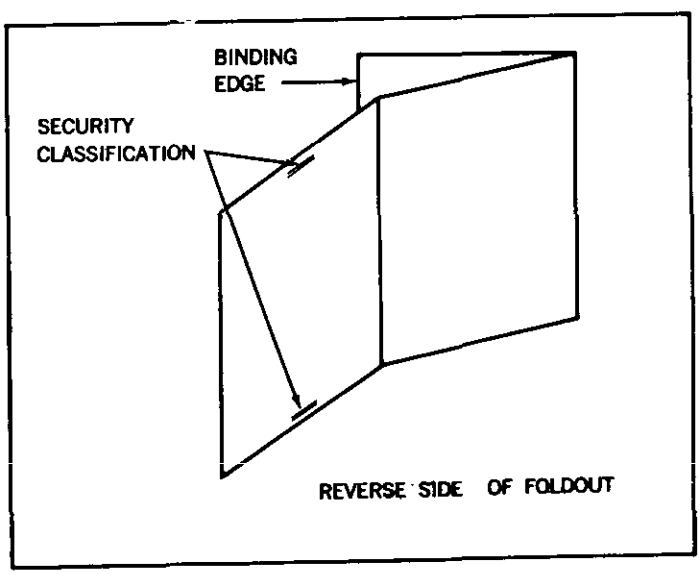


Figure 12. Makeup of Foldover Page (Par. 3.3.2.5 and 3.1.5.2.1b)

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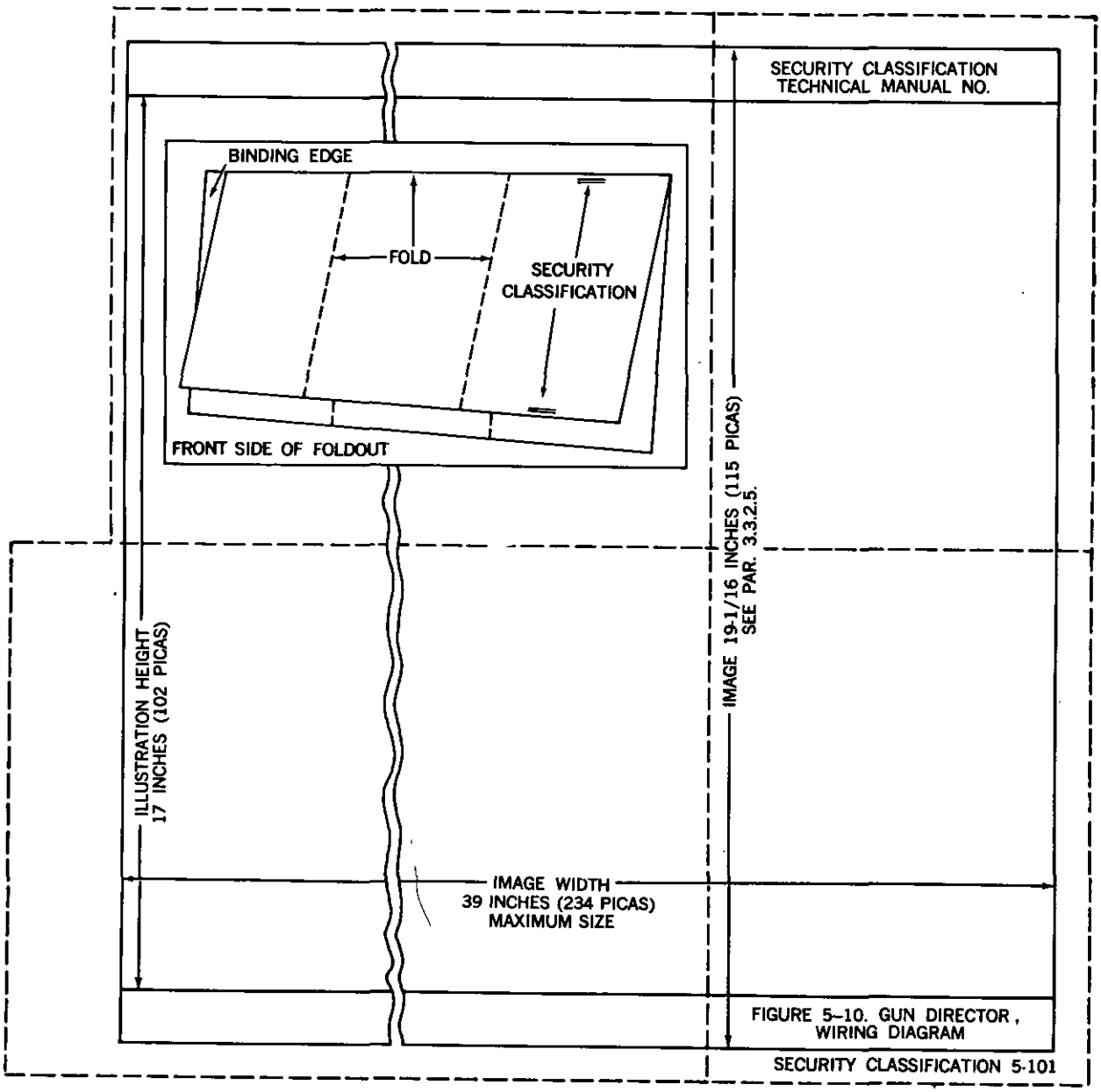


Figure 13. Makeup of Foldover Page with Horizontal Fold. (Par. 3.3.2.5.1)

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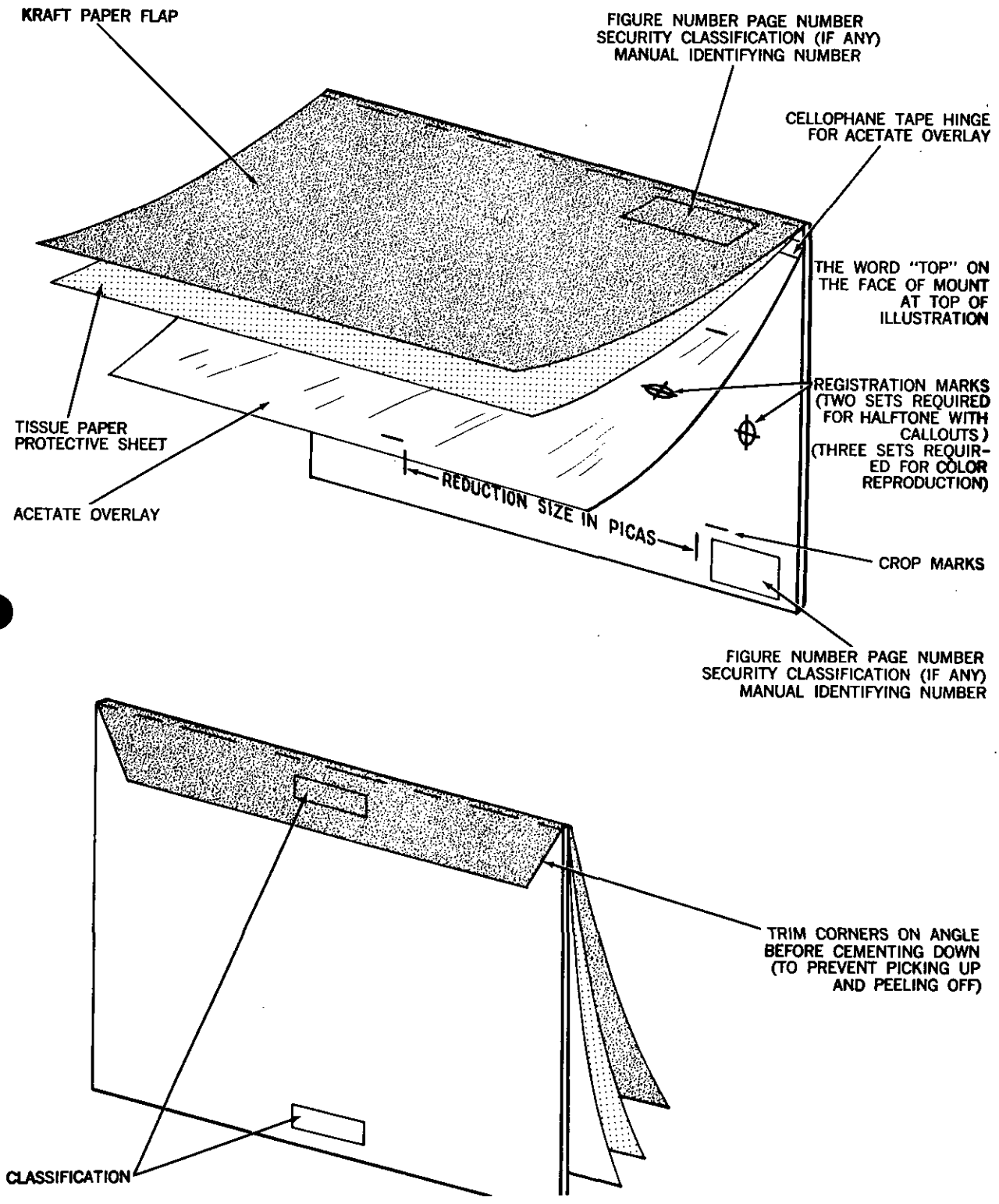


Figure 14. Identification, Marking, and Protective Coverings for Continuous Tone Artwork. (Par. 3.3.2.2.4)

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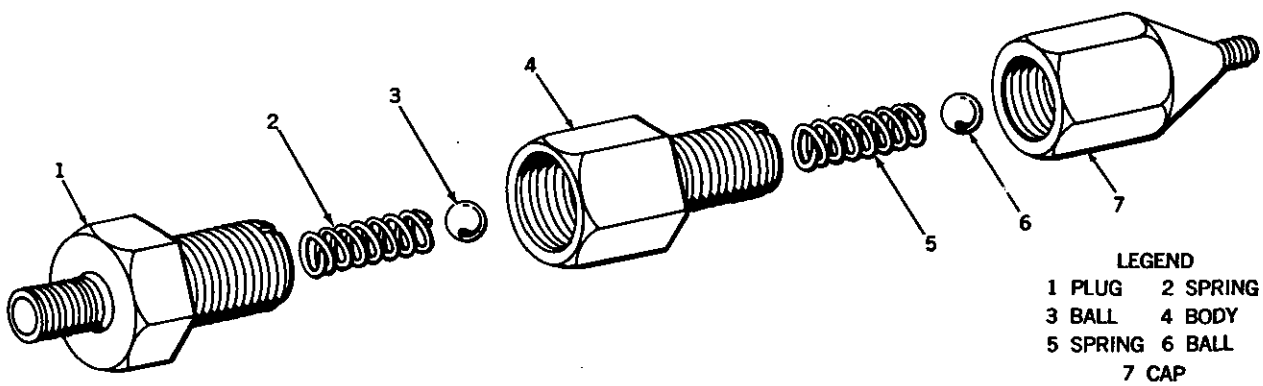


Figure 15. Sample Illustration Showing Use of Callouts. (Par. 3.3.2.3.4.1)

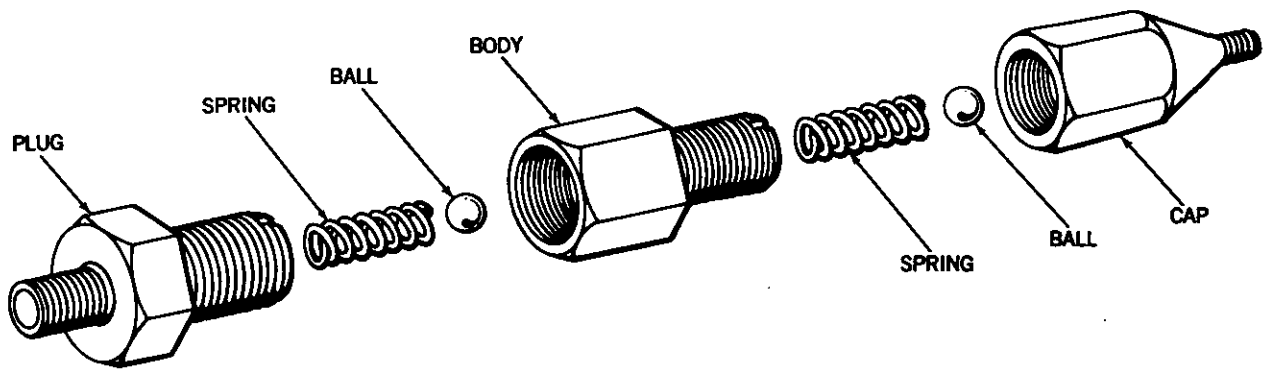


Figure 16. Sample Illustration Showing Use of Brief Nomenclature. (Par. 3.3.2.3.4.1)

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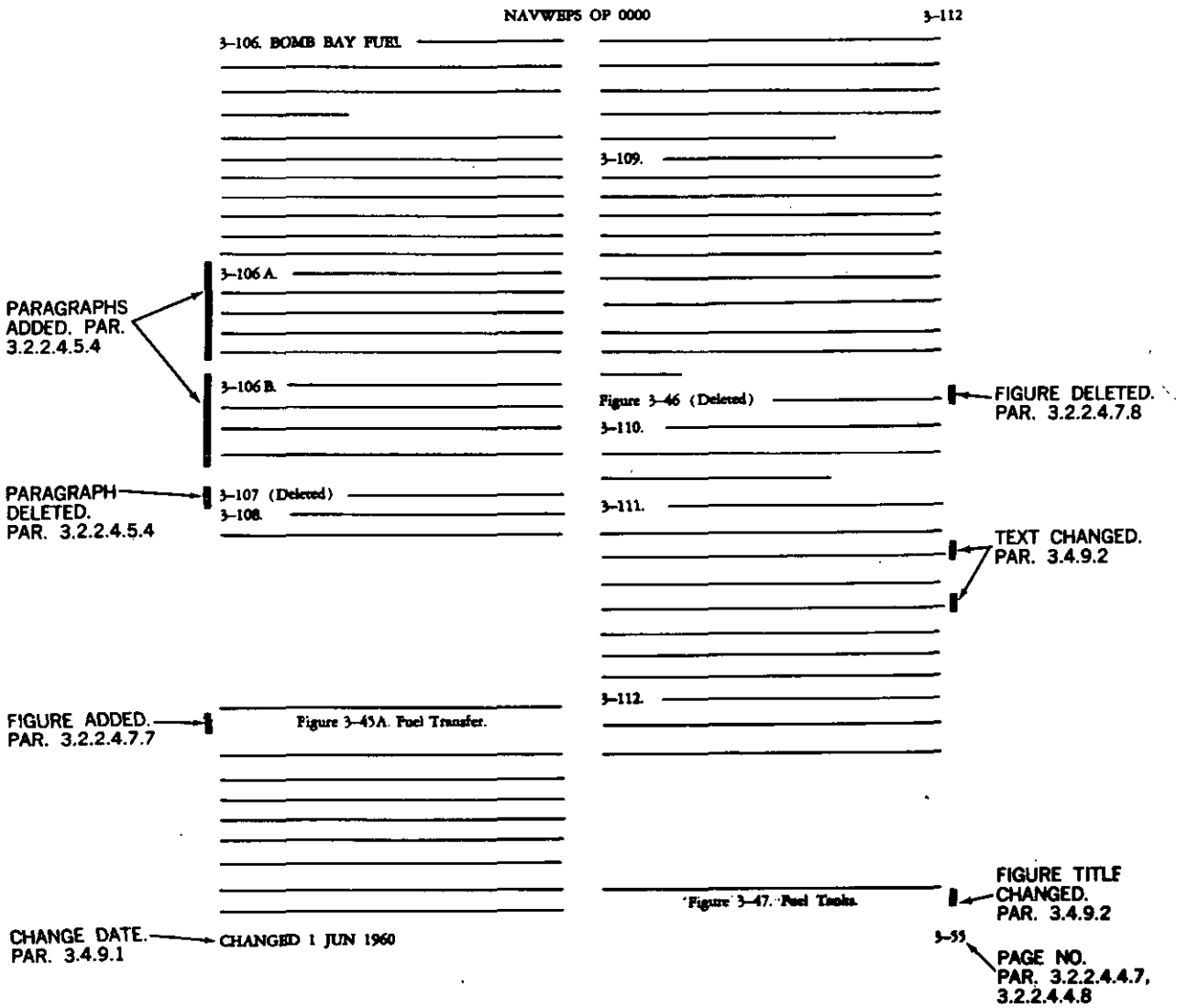


Figure 17. Sample of Changed Page. (Par. 3.4.9)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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